The LNM Institute of Information Technology, Jaipur
Software Development Laboratory
Course Code: CS314
Lab Assignment 1: Project Identification and Initial Project Definition
Project: Curriculum Revision Management System

Team ID	T017
Team Members	Utkarsh, Piyush, Yash, Mehul
Roll Numbers	24UCC096, 24UCC097, 24UCC099, 24UCC100
Development Model	Agile - Iterative and Incremental Development
Date	30 August 2026

1. Project Identification

Item	Details
Project Name	Curriculum Revision Management System
Project Type	Web-based Workflow, Document and Approval Management System
Team ID	T017
Team Members	Utkarsh, Piyush, Yash, Mehul
Roll Numbers	24UCC096, 24UCC097, 24UCC099, 24UCC100
Development Model	Agile - Iterative and Incremental Development
Primary Users	Faculty proposers, Department Curriculum Committee, Board of Studies, Academic Office, Senate/Academic Council and System Administrator
Development Team	Team T017 - student software development team
Course	Software Development Laboratory (CS314)
Project Objective	To provide a controlled, transparent and role-based platform for proposing, reviewing, approving, versioning and implementing curriculum changes across academic units.
Expected Outcome	A working web application that maintains the complete lifecycle and audit trail of every curriculum revision request, from initial proposal to approved implementation.

2. Project Background

Current Problem

Current curriculum revision processes are commonly distributed across email threads, meeting minutes, spreadsheets, shared documents and physical records. A faculty member may submit a proposal in one format, reviewers may provide feedback through several channels, and committees may record decisions separately. As a result, participants cannot easily determine the latest proposal version, pending action, decision rationale or implementation status.

Proposed Solution

The proposed Curriculum Revision Management System will centralize this process. It will provide structured revision proposals, role-based review workflows, stakeholder feedback, committee decisions, version history, notifications, implementation tracking and a complete audit trail. The platform will support transparency and accountability without replacing the academic authority of curriculum committees.

3. Project Vision

To create a reliable and transparent digital platform through which every authorized stakeholder can propose, review, decide upon and track curriculum revisions using one authoritative record from initial idea to academic implementation.

4. Problem Statement

Curriculum revisions involve multiple stakeholders, documents, review stages and formal approvals. When these activities are handled through disconnected files and communication channels, proposals may be delayed, feedback can be overlooked, outdated versions may be reviewed, and decision ownership becomes unclear. The institution therefore needs a role-based software system that preserves proposal context, enforces the approved workflow, records every revision and decision, and provides timely visibility into implementation progress.

5. Project Objectives

The initial objectives are to:

1. Provide authenticated, role-based access for proposers, reviewers, committee members, academic administrators and system administrators.
2. Enable faculty members and departments to create structured proposals for adding, modifying or discontinuing courses, regulations and programme components.
3. Capture the academic rationale, affected programmes, prerequisites, credits, syllabus changes, resource implications and proposed implementation session for each revision.
4. Route proposals through configurable review and approval stages while clearly showing the current owner, status and pending action.
5. Collect stakeholder comments, recommendations, requested changes and supporting documents without losing their context.
6. Maintain immutable version history and allow authorized users to compare successive versions of a proposal.
7. Record committee meetings, decisions, conditions, voting or recommendation summaries and final approval or rejection reasons.
8. Notify relevant stakeholders when a proposal is submitted, returned, revised, scheduled, approved, rejected or awaiting action.
9. Track approved changes through implementation milestones and maintain evidence that academic records have been updated.
10. Provide search, filters, dashboards and auditable reports for pending proposals, decision turnaround and revision history.

6. Stakeholders

Stakeholder	Interest / Role
Faculty Proposer	Creates curriculum revision proposals, supplies academic justification, responds to feedback and submits revised versions.
Programme Coordinator / HOD	Performs departmental review, checks completeness and recommends, returns or forwards proposals.
Department Curriculum Committee / Board of Studies	Reviews academic merit, records feedback and recommendations, and makes committee-level decisions.
Students and Alumni	May provide invited stakeholder feedback on curriculum relevance and learning experience when permitted.
Industry / External Expert	Provides invited feedback about professional relevance, emerging skills and employability requirements.
Academic Office	Checks policy compliance, schedules formal review, maintains official records and tracks implementation.
Senate / Academic Council	Provides final institutional approval or rejection for revisions requiring the highest authority.
System Administrator	Manages users, roles, workflow configuration, master data, access control and system availability.
Development Team T017	Elicits requirements and designs, implements, tests, documents and maintains the software prototype.

7. Initial Scope

The Curriculum Revision Management System will initially cover:

- Secure login and role-based access control.
- Academic programme, department, course and committee master data required by proposals.
- Proposal creation for new courses, course modifications, course discontinuation and programme-structure changes.
- Structured proposal fields, supporting-document upload and draft saving.
- Submission, completeness checking, review, return-for-revision, recommendation, approval and rejection workflow.
- Threaded stakeholder feedback and proposer responses.
- Version history with timestamps, authorship and basic comparison of changed proposal fields.
- Committee meeting reference, decision, conditions and reason recording.
- Email or in-application notifications and pending-action reminders.
- Implementation-status tracking after approval.
- Search, filters, status dashboard, proposal timeline and audit reports.

Initially Out of Scope

To keep the project manageable, the initial release excludes:

- Direct write-back to the institute ERP, website, or course-registration database - requires production API credentials and a formal integration agreement.
- Institutional single sign-on (SSO) - will be simulated via a test authentication adapter until IT grants access.
- Cross-university proposal exchange - requires an inter-institutional data-sharing policy that does not yet exist.
- External government/regulatory approval workflows - outside the institute's internal process and not required for the initial demonstration.

8. Initial Assumptions

- Users will have reliable internet access and an institute-supported web browser.
- Authorized users will receive valid institutional accounts and one or more approved roles.
- The institute will provide an agreed curriculum revision workflow, proposal templates and role hierarchy for configuration.
- Committee decisions entered into the system will be authorized and verified by the designated Academic Office or committee secretary.
- The initial prototype will use representative sample data instead of complete confidential institutional records.
- Uploaded documents will follow supported formats and configured file-size limits.
- A centralized relational database will be available for proposal, workflow and audit data.
- Stakeholders will review notifications and perform assigned actions within the expected academic schedule.

9. Initial Constraints

- The project must be completed within the CS314 laboratory schedule by a student development team.
- Access to real confidential curriculum proposals, committee records and institutional identity systems may be limited.
- The first release must use a manageable workflow and cannot model every exceptional academic policy case.
- Requirements may evolve after consultations with faculty, committee members and the Academic Office.
- The system must protect unpublished proposals, reviewer comments and decision records through role-based authorization.
- Development and deployment will rely on the computing infrastructure and software services available to the team.
- External email, institutional single sign-on and ERP integrations may need to be demonstrated through test adapters.
- Testing with every stakeholder category may not be possible; representative users and test accounts will be used.

10. Initial Success Criteria

The project will be considered successful if the team can demonstrate that:

- an authorized faculty user can create, save, edit and submit a complete curriculum revision proposal;
- the submitted proposal follows the configured review sequence and is visible only to users with appropriate roles;
- reviewers can provide feedback and the proposer can submit a revised version without overwriting earlier versions;
- authorized committee users can record a recommendation, approval, conditional approval, return or rejection with reasons;
- the system displays an accurate timeline containing submissions, feedback, versions, decisions and responsible actors;
- approved proposals can be assigned implementation milestones and tracked to a completed or delayed status;
- search, filters and dashboards correctly report proposals by department, type, stage, owner and implementation status;
- unauthorized users are prevented from viewing or modifying restricted proposals and committee decisions;
- all major workflows pass documented functional tests and the repository contains consistent requirements, design and test artifacts;
- the project is maintained in the team GitHub repository with meaningful commits and contributions from all team members.

AI USAGE POLICY - STUDENT DECLARATION

AI tools may be used as supporting tools for this assignment, but the submitted work must represent the team's own understanding and contribution.

Item	Details
AI Tool Used	ChatGPT
Purpose of AI Use	AI tools were used for language refinement, content organization, and completeness checking while all project decisions and final content were reviewed and prepared by the team.
Team ID	T017
Team Member Names	Utkarsh, Piyush, Yash, Mehul
Date	30 Aug 2026
Team Declaration	We confirm that the information provided above is accurate and that we accept responsibility for the submitted work.

Item	Details
Team Representative Signature / Confirmation	YASH GUPTA

Submission Note: This part establishes the initial project baseline for subsequent requirements engineering, SRS preparation, UML modelling, system design, implementation and testing activities.

The LNM Institute of Information Technology, Jaipur
Software Development Laboratory
Course Code: CS314
Lab Assignment 2: Requirements Specification
Initial SRS (v1.0) - Curriculum Revision Management System

Team ID	T017
Team Members	Utkarsh, Piyush, Yash, Mehul
Roll Numbers	24UCC096, 24UCC097, 24UCC099, 24UCC100
Development Model	Agile - Iterative and Incremental Development
Date	6 September 2026

11. User Classes

11.1 Faculty Proposer

Faculty Proposers initiate curriculum revision requests and remain responsible for the academic content of their proposals.

They shall be able to:

- authenticate and access role-permitted functions;
- create proposals for new courses, course modifications, course discontinuation and programme-structure changes;
- enter academic rationale, affected programmes, prerequisites, credits, syllabus changes, resource implications and proposed implementation session;
- save proposals as drafts;
- upload supporting documents;
- submit complete proposals into the configured review workflow;
- view feedback and requested changes;
- create revised versions without overwriting earlier versions;
- track proposal status, current owner and pending action;
- view notifications and the proposal timeline.

11.2 Programme Coordinator / HOD

The Programme Coordinator / HOD performs department-level scrutiny before a proposal progresses to later academic review stages.

They shall be able to:

- authenticate;
- view proposals assigned to their department;
- perform completeness and departmental review;
- record comments and recommendations;
- return a proposal for revision;
- recommend or forward a proposal to the next configured stage;
- view current and earlier proposal versions;
- view pending actions and relevant notifications.

11.3 Department Curriculum Committee / Board of Studies Reviewer

Committee reviewers examine academic merit and record formal review outcomes at committee level.

They shall be able to:

- authenticate;
- access proposals scheduled or assigned to their committee;
- review proposal content and supporting documents;
- record comments, recommendations and requested changes;
- view version history and compare successive versions;
- associate review activity with a committee meeting where applicable;
- record recommendation, conditional approval, return or rejection decisions permitted by their role.

11.4 Invited Stakeholder

This class includes Students, Alumni, Industry representatives and External Experts who are invited to provide feedback.

They shall be able to:

- access only the proposal material explicitly shared with them;

- submit feedback within the permitted invitation period;
- view their own submitted feedback and proposer/reviewer responses when permitted.

Invited Stakeholders shall not obtain committee approval authority.

11.5 Academic Office

The Academic Office manages policy compliance, formal review coordination, official records and post-approval implementation tracking.

They shall be able to:

- authenticate;
- verify policy and process compliance;
- view and manage proposals requiring Academic Office action;
- schedule or reference formal review meetings;
- maintain official decision records;
- send or review pending-action notifications;
- create and update implementation milestones after approval;
- upload or reference implementation evidence;
- search proposals and generate auditable reports.

11.6 Senate / Academic Council

The Senate / Academic Council provides the final institutional decision for revisions that require the highest level of authority.

They shall be able to:

- authenticate;
- view the formal proposal package and preceding recommendations;
- record final approval or rejection where the configured workflow requires their authority;
- record reasons, conditions or decision notes;
- view the final proposal version and decision history.

11.7 System Administrator

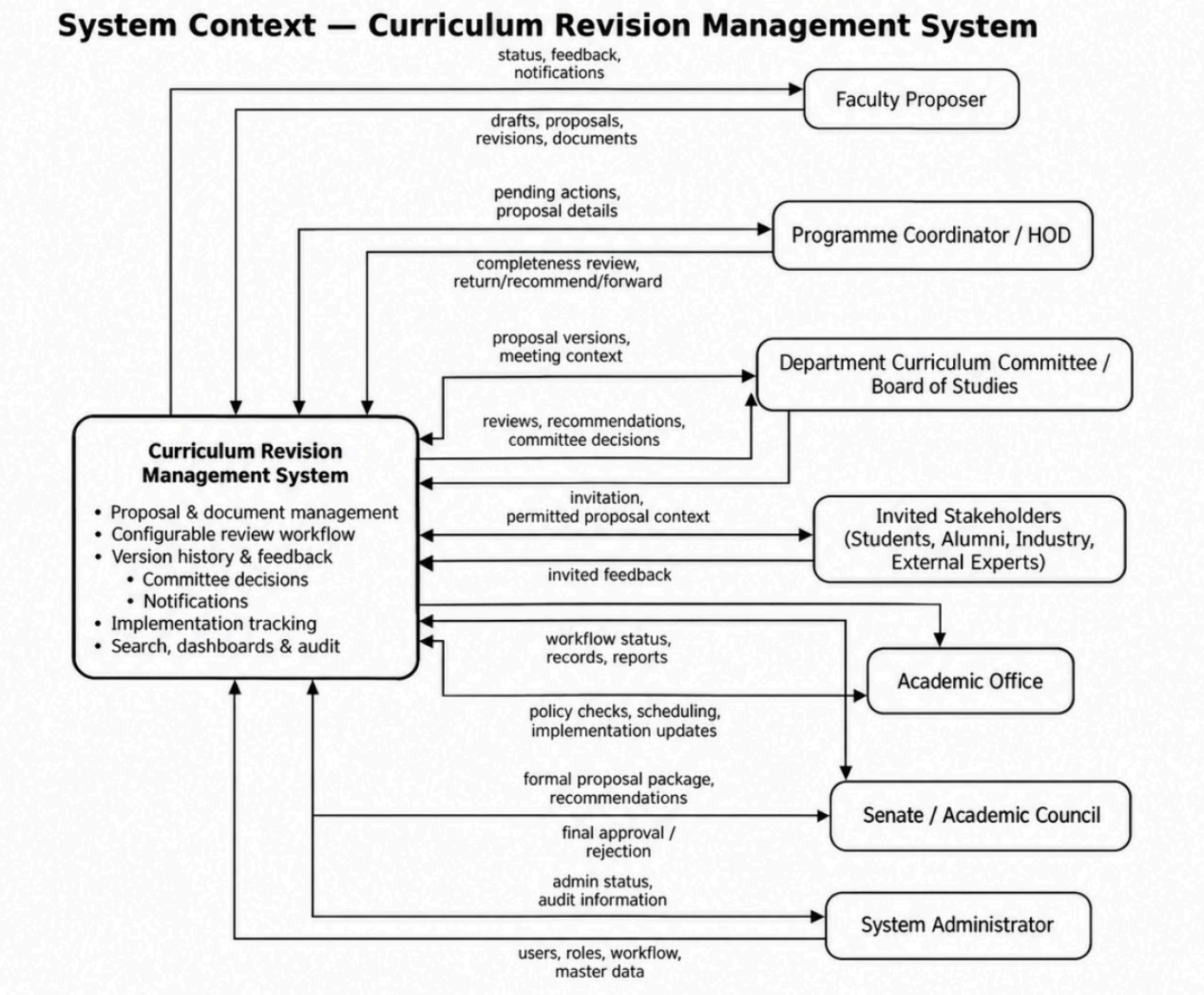
The System Administrator manages technical configuration and access control.

They shall be able to:

- authenticate;
- create, activate, deactivate and update user accounts;
- assign one or more approved roles to users;
- manage departments, programmes, courses and committee master data;
- configure the permitted workflow stages and role-to-stage mappings;
- manage access-control configuration;
- view system-level audit information required for administration.

12. System Context

The Curriculum Revision Management System acts as the authoritative workflow and record-management platform between academic proposers, reviewers, decision-making bodies, invited stakeholders and administrators.



Context-Level Information Flows

External Actor	Information Sent to System	Information Received from System
Faculty Proposer	Drafts, proposals, revised versions, supporting documents, responses	Status, feedback, requested changes, notifications
Programme Coordinator / HOD	Completeness review, comments, return/recommend/forward action	Assigned proposals, current version, pending-action list
DCC / Board of Studies	Review comments, recommendations, decisions	Proposal versions, documents, meeting context
Invited Stakeholders	Invited feedback	Invitation and permitted proposal context
Academic Office	Policy checks, formal review scheduling, implementation updates	Workflow status, official records, reports
Senate / Academic Council	Final approval/rejection and decision notes	Formal proposal package and prior recommendations
System Administrator	Users, roles, workflow configuration, master data	Administrative status and audit information

13. Functional Requirements

13.1 Authentication and Role-Based Access

FR-001 - User Login
The system shall allow a registered user to authenticate using valid credentials.

FR-002 - Authentication Failure
The system shall notify a user when authentication fails without revealing whether a specific account, password or role value was correct.

FR-003 - Logout
The system shall allow an authenticated user to terminate the current session.

FR-004 - Role Assignment
The system shall allow a System Administrator to assign one or more approved roles to a user.

FR-005 - Role-Based Dashboard
The system shall present dashboard information and pending actions according to the authenticated user's assigned role or roles.

FR-006 - Role-Based Authorization
The system shall prevent a user from viewing or modifying proposal data, decisions, comments or administrative functions for which the user lacks permission.

13.2 Academic Master Data and Workflow Configuration

FR-007 - Manage Academic Master Data
The system shall allow a System Administrator to create and update department, programme, course and committee master data required by curriculum proposals.

FR-008 - Configure Review Workflow
The system shall allow a System Administrator to configure the ordered review stages, permitted actions and responsible role for each stage used by the initial prototype.

13.3 Proposal Creation and Draft Management

FR-009 - Create Curriculum Revision Proposal
The system shall allow an authorized Faculty Proposer to create a new curriculum revision proposal.

FR-010 - Select Proposal Type
The system shall require the proposer to classify a proposal as a new course, course modification, course discontinuation or programme-structure change.

FR-011 - Capture Structured Proposal Details
The system shall capture the proposal title, academic rationale, affected programme(s), prerequisites, credits, syllabus changes, resource implications and proposed implementation session where applicable.

FR-012 - Save Draft Proposal
The system shall allow a proposer to save an incomplete proposal as a draft without submitting it for review.

FR-013 - Upload Supporting Documents
The system shall allow an authorized proposer to attach supported documents to a draft or returned proposal subject to configured file type and size limits.

FR-014 - Edit Draft or Returned Proposal
The system shall allow the proposal owner to edit a proposal only while it is in Draft or Returned for Revision status.

FR-015 - Validate Proposal Completeness
The system shall validate all mandatory fields and required supporting information before allowing submission.

FR-016 - Submit Proposal
The system shall allow the proposal owner to submit a complete proposal into the configured review workflow.

FR-017 - Display Proposal Status and Owner
The system shall display the current proposal status, current workflow stage, responsible role and pending action to authorized users.

13.4 Review, Feedback and Workflow Routing

FR-018 - Route Submitted Proposal
The system shall route a submitted proposal to the first configured review stage and assign the pending action to the role responsible for that stage.

FR-019 - Review Assigned Proposal
The system shall allow an authorized reviewer to open and review a proposal assigned to the reviewer's role or committee.

FR-020 - Record Review Comment
The system shall allow an authorized reviewer to add a contextual comment, recommendation or requested change to the proposal.

FR-021 - Return for Revision
The system shall allow an authorized reviewer to return a proposal to its proposer for revision with a recorded reason.

FR-022 - Recommend or Forward Proposal
The system shall allow an authorized reviewer to recommend or forward a proposal to the next configured workflow stage.

FR-023 - Resubmit Revised Proposal
The system shall allow the proposer to resubmit a returned proposal after creating a new version that addresses requested changes.

FR-024 - Enforce Workflow Sequence
The system shall prevent a proposal from bypassing a mandatory configured review stage unless the workflow configuration explicitly permits that transition.

13.5 Version History and Comparison

FR-025 - Create Version Snapshot
The system shall create an immutable version snapshot whenever a proposal is submitted or resubmitted.

FR-026 - View Version History
The system shall allow authorized users to view all saved submission versions with version number, author and timestamp.

FR-027 - Compare Proposal Versions
The system shall allow authorized users to compare two successive proposal versions and identify changed structured fields.

13.6 Invited Stakeholder Feedback

FR-028 - Invite Stakeholder Feedback
The system shall allow an authorized academic reviewer or Academic Office user to invite a permitted stakeholder to provide feedback on a proposal.

FR-029 - Submit Invited Feedback
The system shall allow an invited stakeholder to submit feedback only for a proposal for which a valid invitation exists.

FR-030 - Preserve Feedback Context
The system shall associate each feedback item and authorized response with the relevant proposal, version, author and timestamp.

13.7 Committee Meetings and Decisions

FR-031 - Record Meeting Reference
The system shall allow an authorized committee or Academic Office user to associate a proposal with a committee meeting reference and meeting date.

FR-032 - Record Committee Decision
The system shall allow an authorized committee user to record a permitted decision such as recommendation, conditional approval, return or rejection.

FR-033 - Record Decision Reason or Conditions
The system shall record the reason, conditions or decision notes required for a formal committee decision.

FR-034 - Record Recommendation or Voting Summary
The system shall allow an authorized committee user to record a recommendation or voting summary when such information is part of the configured process.

FR-035 - Record Final Institutional Decision
The system shall allow the Senate / Academic Council role to record final approval or rejection for proposals whose configured workflow requires that authority.

13.8 Notifications and Pending Actions

FR-036 - Event Notification
The system shall create an in-application notification for relevant stakeholders when a proposal is submitted, returned, revised, scheduled, approved, rejected or assigned for action.

FR-037 - Pending-Action Reminder
The system shall identify workflow actions that remain pending beyond a configurable reminder interval and generate a reminder for the responsible role.

13.9 Implementation Tracking

FR-038 - Create Implementation Milestone
The system shall allow an authorized Academic Office user to create implementation milestones for an approved proposal.

FR-039 - Update Implementation Status
The system shall allow an authorized Academic Office user to update a milestone or proposal implementation status.

FR-040 - Record Implementation Evidence
The system shall allow an authorized Academic Office user to attach or reference evidence that an approved curriculum change has been implemented.

FR-041 - Track Completion or Delay
The system shall display whether implementation is Not Started, In Progress, Completed or Delayed, together with the latest recorded milestone information.

13.10 Search, Dashboard, Timeline and Audit Reporting

FR-042 - Search Proposals
The system shall allow an authorized user to search proposals by proposal ID, title or keyword.

FR-043 - Filter Proposals
The system shall allow an authorized user to filter proposals by department, proposal type, workflow stage, current owner and implementation status as permitted by access rights.

FR-044 - Status Dashboard
The system shall display role-scoped counts of proposals by relevant status and pending-action category.

FR-045 - Proposal Timeline
The system shall display a chronological timeline of major proposal events including submissions, feedback, version creation, review actions, decisions and implementation updates.

FR-046 - Audit Report
The system shall allow an authorized Academic Office or System Administrator user to view or generate an audit-oriented report of proposal actions for a selected proposal or period.

FR-047 - Record Audit Event
The system shall record significant security, workflow, decision and administrative actions with actor, action, affected object and timestamp.

FR-048 - View Audit Trail
The system shall allow only authorized roles to view the audit trail relevant to their administrative responsibility.

14. Business Rules

ID	Business Rule
BR-001	Only a System Administrator may create or modify user accounts, role assignments, workflow configuration and core academic master data.
BR-002	A user may hold one or more approved roles, but the permissions exercised in a session shall be limited to those assigned roles.
BR-003	A proposal owner may modify proposal content only while the proposal is in Draft or Returned for Revision status.
BR-004	A proposal cannot be submitted unless all mandatory structured fields and required supporting information for its proposal type are complete.
BR-005	Each submission or resubmission shall create a new immutable proposal version.
BR-006	Earlier submitted versions shall remain viewable to authorized users and shall not be overwritten by later revisions.
BR-007	A submitted proposal shall follow the configured review sequence and shall not skip a mandatory stage.
BR-008	A Return for Revision or Rejection action shall include a recorded reason.
BR-009	A Conditional Approval action shall include the applicable condition(s).

ID	Business Rule
BR-010	Only users assigned to the responsible role or committee for the current workflow stage may perform the stage's formal review action.
BR-011	Invited Students, Alumni, Industry representatives and External Experts may provide feedback only when explicitly invited and shall not receive approval authority.
BR-012	A final institutional approval or rejection may be recorded by Senate / Academic Council only when the configured workflow requires that authority.
BR-013	Implementation milestones may be created only after the proposal has reached an approved state.
BR-014	Committee decisions entered into the system shall be attributable to an authorized committee user and may be verified by the designated Academic Office or committee secretary.
BR-015	Unpublished proposals, reviewer comments and decision records shall be visible only to roles permitted by the workflow and access-control configuration.
BR-016	The proposal's displayed current owner and pending action shall be derived from the active workflow stage.
BR-017	Audit events for formal submissions, review actions, role changes and decisions shall not be editable by ordinary users.
BR-018	Uploaded supporting documents shall comply with configured supported formats and file-size limits.

15. Non-Functional Requirements

15.1 Performance

NFR-001: Under the expected prototype workload, at least 95% of normal authenticated page and CRUD requests should complete within 3 seconds, excluding file upload time.

NFR-002: Proposal search and filter operations should return the first result page within 3 seconds for the representative dataset used during acceptance testing.

NFR-003: The system should support at least 50 concurrent authenticated prototype users without functional failure.

15.2 Security

NFR-004: Protected functions and proposal data shall require authentication.

NFR-005: Authorization shall be enforced on the server side for every protected operation, not only by hiding user-interface controls.

NFR-006: Passwords shall be stored using a secure one-way password hashing mechanism and shall never be stored in plain text.

NFR-007: Session identifiers and authentication tokens shall not be exposed in application logs.

NFR-008: A user shall be denied access to proposal, feedback, decision and administrative data outside the user's permitted role and workflow scope.

NFR-009: Production-style deployment shall use encrypted HTTPS transport where hosting infrastructure supports it.

15.3 Usability

NFR-010: Major modules shall use consistent navigation, terminology and status labels.

NFR-011: Validation and error messages shall identify the field or action requiring correction in understandable language.

NFR-012: The user interface shall clearly display proposal status, current owner and pending action on the proposal details view.

NFR-013: The web interface should remain usable on common laptop and tablet screen sizes without horizontal scrolling on primary forms.

15.4 Reliability

NFR-014: A successfully confirmed proposal submission, review action, decision or version creation shall remain persisted after page refresh or user re-login.

NFR-015: Failure during document upload or workflow action shall be reported to the user and shall not silently create an incomplete formal action.

NFR-016: Version history shall preserve all previously submitted versions even when a newer version is created.

15.5 Availability

NFR-017: The hosted prototype should be available during scheduled laboratory demonstrations and agreed testing periods except during announced maintenance.

15.6 Maintainability

NFR-018: The backend shall separate major concerns including authentication, proposal management, workflow, feedback, decisions, notifications, implementation tracking and audit logging.

NFR-019: Source code and requirements documentation shall be maintained using Git with meaningful commits.

NFR-020: Requirement IDs, use-case IDs and user-story IDs shall remain stable within SRS v1.x unless a controlled change explicitly retires an item.

15.7 Scalability

NFR-021: The design should allow additional departments, programmes, courses, committees and workflow stages to be added through data/configuration changes rather than fundamental redesign.

15.8 Privacy

NFR-022: Unpublished proposal content and reviewer comments shall not be made publicly accessible.

NFR-023: An invited stakeholder shall receive access only to the proposal information necessary for the specific invitation.

15.9 Compatibility

NFR-024: The web application should operate correctly on current stable versions of Google Chrome, Microsoft Edge and Mozilla Firefox used in the institute environment.

15.10 Auditability

NFR-025: Each auditable event shall record at minimum the actor, action, affected object, result and timestamp.

NFR-026: Audit and version-history timestamps shall be generated by the server to maintain consistent ordering of events.

16. Data Requirements

The initial logical data model shall include the following major entities.

16.1 User

Important attributes:

- User ID
- Name
- Institutional email
- Password hash / authentication reference
- Account status
- Department
- Created date
- Last updated date

16.2 Role

Important attributes:

- Role ID
- Role name
- Role description
- Permission set

16.3 UserRole

Important attributes:

- User ID
- Role ID
- Assigned by
- Assigned date
- Active status

16.4 Department

Important attributes:

- Department ID
- Department name
- Department code
- Active status

16.5 Programme

Important attributes:

- Programme ID
- Programme name
- Programme code
- Department ID
- Degree level
- Active status

16.6 Course

Important attributes:

- Course ID
- Course code
- Course title
- Credits
- Department ID
- Current status

16.7 Committee

Important attributes:

- Committee ID
- Committee name
- Committee type
- Department / institutional scope
- Active status

16.8 CurriculumProposal

Important attributes:

- Proposal ID
- Proposal type
- Title
- Owner / proposer ID
- Department ID
- Affected programme(s)
- Current status
- Current workflow stage
- Current responsible role
- Proposed implementation session
- Created date
- Last updated date

16.9 ProposalVersion

Important attributes:

- Version ID
- Proposal ID
- Version number
- Academic rationale
- Prerequisites
- Credits
- Syllabus-change description
- Resource implications

- Structured proposal fields
- Created by
- Created timestamp
- Submission timestamp

16.10 SupportingDocument

Important attributes:

- Document ID
- Proposal ID
- Version ID
- File name
- File type
- File reference / storage path
- Uploaded by
- Uploaded timestamp

16.11 WorkflowDefinition

Important attributes:

- Workflow ID
- Workflow name
- Proposal type / scope
- Active status
- Version

16.12 WorkflowStage

Important attributes:

- Stage ID
- Workflow ID
- Stage sequence
- Stage name
- Responsible role
- Permitted actions
- Mandatory flag

16.13 ReviewAction

Important attributes:

- Review Action ID
- Proposal ID
- Proposal Version ID
- Stage ID
- Reviewer ID
- Action type
- Comment / reason
- Action timestamp

16.14 StakeholderInvitation

Important attributes:

- Invitation ID
- Proposal ID
- Invitee identity / email
- Stakeholder category
- Invited by
- Valid-from date
- Expiry date
- Status

16.15 Feedback

Important attributes:

- Feedback ID

- Proposal ID
- Proposal Version ID
- Author / invitation ID
- Parent feedback ID (for threaded response)
- Feedback text
- Created timestamp

16.16 CommitteeMeeting

Important attributes:

- Meeting ID
- Committee ID
- Meeting reference / number
- Meeting date
- Notes / record reference

16.17 Decision

Important attributes:

- Decision ID
- Proposal ID
- Proposal Version ID
- Meeting ID
- Decision type
- Decision reason
- Conditions
- Recommendation / voting summary
- Recorded by
- Decision timestamp

16.18 Notification

Important attributes:

- Notification ID
- Recipient user / role
- Proposal ID
- Event type
- Message
- Created timestamp
- Read status

16.19 ImplementationMilestone

Important attributes:

- Milestone ID
- Proposal ID
- Milestone title
- Target date
- Completion date
- Status
- Evidence reference
- Updated by
- Updated timestamp

16.20 AuditEvent

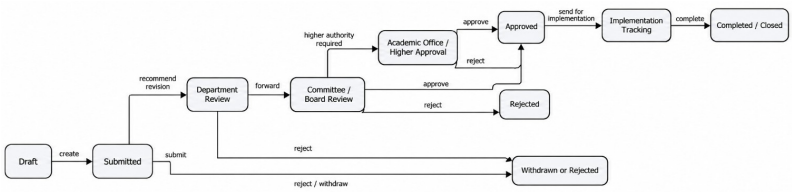
Important attributes:

- Audit Event ID
- Actor ID
- Action
- Object type
- Object ID
- Result
- Timestamp

- Additional metadata

17. Proposal Lifecycle

The curriculum proposal is treated as a controlled record whose state changes only through permitted workflow actions. A typical proposal lifecycle for the initial release is shown below.



The exact sequence can vary according to configured proposal type and authority. However, the following principles apply:

- Draft proposals remain editable by their owner.
- Submission creates an immutable version and transfers responsibility to the first configured review stage.
- A reviewer may return the proposal for revision; resubmission creates a new version while retaining earlier versions.
- Committee review may result in recommendation, conditional approval, return or rejection according to authority.
- Proposals requiring the highest authority are routed to Senate / Academic Council after the preceding recommendation stages.
- Approved proposals enter implementation tracking; rejected proposals become final unless a later controlled change process reopens them.
- Every formal state transition is recorded in the timeline and audit trail.

18. Major Use Cases

Use Case ID	Use Case	Primary Actor
UC-001	Authenticate User	All registered users
UC-002	Manage Users and Roles	System Administrator
UC-003	Manage Academic Master Data	System Administrator
UC-004	Create / Save Draft Proposal	Faculty Proposer
UC-005	Submit Proposal	Faculty Proposer
UC-006	Upload Supporting Documents	Faculty Proposer
UC-007	View Proposal Status and Pending Action	Authorized Stakeholder
UC-008	Perform Department Completeness Review	Programme Coordinator / HOD
UC-009	Review Proposal and Record Feedback	Authorized Reviewer / Committee Member
UC-010	Return Proposal for Revision	Authorized Reviewer
UC-011	Submit Revised Proposal Version	Faculty Proposer
UC-012	View / Compare Proposal Versions	Authorized Reviewer / Proposer
UC-013	Invite Stakeholder Feedback	Authorized Reviewer / Academic Office
UC-014	Submit Invited Feedback	Invited Stakeholder
UC-015	Record Committee Decision	Authorized Committee User
UC-016	Record Final Institutional Decision	Senate / Academic Council
UC-017	Configure Review Workflow	System Administrator
UC-018	View Notifications	All registered users
UC-019	Track Approved Change Implementation	Academic Office
UC-020	Search and Filter Proposals	Authorized Stakeholder
UC-021	View Role-Scoped Dashboard	Authorized Stakeholder
UC-022	View Proposal Timeline	Authorized Stakeholder
UC-023	View Audit Trail / Audit Report	Academic Office / System Administrator
UC-024	Record Committee Meeting Reference	Committee User / Academic Office
UC-025	View Pending Actions	Reviewer / Academic Office / Senate

19. Detailed Use Cases

19.1 UC-005 - Submit Proposal

Primary Actor: Faculty Proposer

Preconditions

- The Faculty Proposer is authenticated.
- The proposal exists and is owned by the Faculty Proposer.
- The proposal is in Draft or Returned for Revision status.
- The applicable workflow is configured and active.

Main Flow

1. The proposer opens the proposal.
2. The system displays the current editable proposal data and supporting documents.
3. The proposer selects **Submit**.
4. The system validates all mandatory fields and required information for the selected proposal type.
5. The system creates an immutable proposal-version snapshot.
6. The system changes the proposal from an editable state to Submitted / Under Review.
7. The system determines the first configured review stage and responsible role.
8. The system assigns the pending action to that stage.
9. The system records submission and version events in the timeline and audit trail.
10. The system generates relevant in-application notifications.
11. The system displays confirmation and the updated current status.

Alternative / Exception Flows

- **A1 - Mandatory data missing:** If required fields are incomplete, the system does not submit the proposal and identifies the missing or invalid fields.
- **A2 - Required document invalid:** If a supporting document violates a configured file rule, the system rejects that file and keeps the proposal editable.
- **A3 - Workflow unavailable:** If no active workflow applies, the system prevents submission and reports that workflow configuration is required.

- **A4 - Unauthorized user:** If the authenticated user does not own the editable proposal, the system denies the submission action.

Postconditions

- A new immutable submitted proposal version exists.
- The proposal is assigned to the first configured review stage.
- The proposer can no longer alter the submitted version.
- The new status, pending action, timeline event and audit event are visible to authorized users.

19.2 UC-009 - Review Proposal and Record Feedback

Primary Actor: Authorized Reviewer / Committee Member

Preconditions

- The reviewer is authenticated.
- The reviewer has a role permitted to act at the proposal's current workflow stage.
- The proposal has at least one submitted version.
- The proposal is assigned to the reviewer's role or committee.

Main Flow

1. The reviewer opens the pending-action list.
2. The system displays proposals assigned to the reviewer's role.
3. The reviewer selects a proposal.
4. The system displays the current proposal version, supporting documents, previous review context and relevant version history.
5. The reviewer evaluates the proposal.
6. The reviewer records a comment, recommendation or requested change.
7. The reviewer selects a permitted workflow action, such as Recommend/Forward or Return for Revision.
8. The system validates that the selected action is permitted at the current stage.
9. If the action is Return for Revision, the system requires a reason.
10. The system records the review action with reviewer identity and timestamp.
11. The system updates proposal status/current owner according to the workflow.
12. The system creates timeline, audit and notification events.

Alternative / Exception Flows

- **A1 - Reviewer no longer authorized:** The system denies the action if the reviewer's role no longer has permission at the current stage.
- **A2 - Stale proposal stage:** If another authorized action has already moved the proposal to a different stage, the system prevents the stale action and refreshes the current status.
- **A3 - Missing return reason:** The system does not allow Return for Revision until a reason is entered.
- **A4 - Comment-only review:** The reviewer may save a contextual comment without advancing the workflow if the current stage permits comment-only activity.

Postconditions

- The review action is stored with actor and timestamp.
- The proposal remains at the current stage or moves to the configured next owner depending on the action.
- The proposer or next responsible role receives the relevant notification.
- The event appears in the proposal timeline and audit trail.

19.3 UC-015 - Record Committee Decision

Primary Actor: Authorized Committee User

Preconditions

- The committee user is authenticated.
- The proposal has reached a workflow stage at which the committee is authorized to decide.
- The current submitted version is available to the committee.
- Meeting context is available when required by the configured process.

Main Flow

1. The committee user opens the proposal assigned for decision.
2. The system displays the current version, supporting documents, prior recommendations and relevant feedback.
3. The committee user associates the proposal with a meeting reference if required.
4. The committee user selects a permitted decision: Recommendation, Conditional Approval, Return or Rejection.
5. The committee user enters the required reason, conditions or recommendation/voting summary.

6. The system validates that the decision and supporting fields satisfy the business rules.
7. The system records the decision with the proposal version, actor and timestamp.
8. The system updates the workflow to the configured next stage or final state.
9. The system creates timeline and audit events.
10. The system notifies the proposer, Academic Office and next responsible role as applicable.

Alternative / Exception Flows

- **A1 - Missing reason:** For Return or Rejection, the system prevents saving until a reason is recorded.
- **A2 - Missing conditions:** For Conditional Approval, the system prevents saving until at least one condition is recorded.
- **A3 - Unauthorized decision type:** The system prevents a decision not permitted for the current committee/stage.
- **A4 - Higher authority required:** If the workflow requires Senate / Academic Council approval, the system records the committee outcome as a recommendation and routes the proposal onward rather than marking it finally approved.
- **A5 - Proposal already moved:** If the proposal is no longer at the committee's stage, the system prevents duplicate decision recording.

Postconditions

- The formal committee decision is stored against the relevant proposal version.
- The decision is visible in the authorized proposal history and audit trail.
- The proposal is routed to the next configured stage or reaches the appropriate final state.

20. User Stories

US-001
As a registered user, I want to log in securely so that I can access only the curriculum-revision functions permitted to my role.

US-002
As a Faculty Proposer, I want to create and save a structured curriculum revision proposal as a draft so that I can complete it over multiple sessions.

US-003
As a Faculty Proposer, I want to submit a complete proposal so that it enters the formal academic review workflow.

US-004
As a Faculty Proposer, I want to revise a returned proposal without overwriting earlier submitted versions so that the evolution of the proposal remains traceable.

US-005
As a Programme Coordinator / HOD, I want to review proposal completeness and return, recommend or forward it so that incomplete or unsuitable proposals do not bypass departmental scrutiny.

US-006
As a Committee Reviewer, I want to record contextual comments and recommendations so that academic review reasoning remains attached to the proposal.

US-007
As an invited Student, Alumnus, Industry representative or External Expert, I want to provide feedback on a proposal shared with me so that relevant stakeholder perspectives can inform the revision.

US-008
As an Academic Office user, I want to verify process compliance and coordinate formal review records so that the institute has one authoritative record of the revision process.

US-009
As an authorized Committee Member, I want to record a formal decision with reasons or conditions so that the committee outcome is documented and auditable.

US-010
As a Senate / Academic Council member, I want to record the final institutional decision for proposals requiring my authority so that the formal approval process is completed.

US-011
As an authorized proposer or reviewer, I want to view and compare proposal versions so that I can understand exactly what changed between submissions.

US-012
As an authorized stakeholder, I want to receive notifications and see pending actions so that I know when my response is required.

US-013
As an Academic Office user, I want to track implementation milestones and evidence for approved proposals so that approved curriculum changes are followed through to completion.

US-014
As an authorized stakeholder, I want to search, filter and view the proposal timeline/dashboard so that I can quickly understand proposal status and history.

US-015
As a System Administrator, I want to manage users, roles, academic master data and workflow configuration so that the system reflects the institute's approved structure and access rules.

US-016
As an Academic Office or System Administrator user, I want to view auditable records of major actions so that accountability can be demonstrated during review.

21. Acceptance Criteria

21.1 US-003 - Submit a Complete Proposal

ID	Acceptance Criterion
AC-US003-01	The Faculty Proposer must be authenticated and must own the proposal being submitted.
AC-US003-02	The proposal must be in Draft or Returned for Revision status.
AC-US003-03	The system must prevent submission when mandatory structured fields are incomplete.
AC-US003-04	On successful submission, the system must create a new immutable proposal-version snapshot.
AC-US003-05	The submitted proposal must be assigned to the first configured review stage and responsible role.
AC-US003-06	The submitted version must no longer be editable.
AC-US003-07	The submission must create timeline and audit entries and notify the relevant next actor.

21.2 US-004 - Revise Without Overwriting Earlier Versions

ID	Acceptance Criterion
AC-US004-01	Revision editing must be available only when the proposal has been returned for revision.
AC-US004-02	The proposer must be able to view the reason and contextual feedback associated with the return action.
AC-US004-03	Resubmission must create a new version number rather than modifying the previous submitted version.
AC-US004-04	Earlier submitted versions must remain viewable to authorized users after resubmission.
AC-US004-05	The version history must show author and timestamp for each submitted version.
AC-US004-06	The system must allow authorized users to compare the new version with the immediately preceding submitted version.

21.3 US-009 - Record a Formal Committee Decision

ID	Acceptance Criterion
AC-US009-01	The actor must be authenticated and authorized for the proposal's current committee workflow stage.
AC-US009-02	The decision must be stored against the exact proposal version reviewed by the committee.
AC-US009-03	Return or Rejection must not be saved without a reason.
AC-US009-04	Conditional Approval must not be saved without recorded condition(s).
AC-US009-05	When a higher approval stage is required, a committee recommendation must route the proposal to that stage rather than mark it finally approved.
AC-US009-06	A successful decision must create timeline and audit entries and notify relevant stakeholders.

22. Requirements Prioritization - MoSCoW

Must Have

- Authentication and server-side role-based access control - FR-001 to FR-006
- Academic master data required to create valid proposals - FR-007
- Core structured proposal creation, draft saving, document upload and submission - FR-009 to FR-018
- Department/committee review, comments, return, recommendation and workflow enforcement - FR-019 to FR-024
- Immutable version history - FR-025 to FR-026
- Formal committee and final decision recording - FR-031 to FR-035
- Basic in-application event notifications - FR-036
- Implementation milestone/status tracking - FR-038 to FR-041
- Search, filtering, status dashboard and proposal timeline - FR-042 to FR-045
- Core audit event recording and authorized audit access - FR-047 to FR-048

Should Have

- Configurable workflow stages through administration instead of code changes - FR-008
- Basic field-level comparison between successive proposal versions - FR-027
- Invited stakeholder feedback - FR-028 to FR-030
- Pending-action reminders - FR-037
- Audit-oriented report view - FR-046

Could Have

- Email delivery in addition to mandatory in-application notifications
- Rich visual diff for uploaded documents in addition to structured-field comparison
- Advanced turnaround-time analytics beyond the basic status dashboard
- Downloadable PDF/CSV audit and proposal summary exports
- Calendar-style visualization for committee scheduling

Won't Have in Initial Release

- Direct write-back to institute ERP, website or course-registration database
- Institutional single sign-on (SSO)
- Cross-university proposal exchange
- External government/regulatory approval workflows

23. Requirements Traceability

Initial Requirements Traceability Matrix. Testing references are intentionally marked **TBD** at this stage.

Functional Requirement	User Story	Use Case	Test
FR-001	US-001	UC-001	TBD
FR-002	US-001	UC-001	TBD
FR-003	US-001	UC-001	TBD
FR-004	US-015	UC-002	TBD
FR-005	US-001, US-014	UC-021	TBD
FR-006	US-001, US-015	UC-001, UC-002	TBD
FR-007	US-015	UC-003	TBD
FR-008	US-015	UC-017	TBD
FR-009	US-002	UC-004	TBD
FR-010	US-002	UC-004	TBD
FR-011	US-002	UC-004	TBD
FR-012	US-002	UC-004	TBD
FR-013	US-002	UC-006	TBD
FR-014	US-002, US-004	UC-004, UC-011	TBD
FR-015	US-003	UC-005	TBD
FR-016	US-003	UC-005	TBD
FR-017	US-003, US-014	UC-007	TBD
FR-018	US-003	UC-005	TBD
FR-019	US-005, US-006	UC-008, UC-009	TBD
FR-020	US-006	UC-009	TBD
FR-021	US-005	UC-010	TBD
FR-022	US-005, US-006	UC-009	TBD
FR-023	US-004	UC-011	TBD

Functional Requirement	User Story	Use Case	Test
FR-024	US-005, US-015	UC-009, UC-017	TBD
FR-025	US-004, US-011	UC-011, UC-012	TBD
FR-026	US-011	UC-012	TBD
FR-027	US-011	UC-012	TBD
FR-028	US-006, US-008	UC-013	TBD
FR-029	US-007	UC-014	TBD
FR-030	US-007	UC-014	TBD
FR-031	US-008, US-009	UC-024	TBD
FR-032	US-009	UC-015	TBD
FR-033	US-009	UC-015	TBD
FR-034	US-009	UC-015	TBD
FR-035	US-010	UC-016	TBD
FR-036	US-012	UC-018	TBD
FR-037	US-012	UC-025	TBD
FR-038	US-013	UC-019	TBD
FR-039	US-013	UC-019	TBD
FR-040	US-013	UC-019	TBD
FR-041	US-013	UC-019	TBD
FR-042	US-014	UC-020	TBD
FR-043	US-014	UC-020	TBD
FR-044	US-014	UC-021	TBD
FR-045	US-014	UC-022	TBD
FR-046	US-016	UC-023	TBD
FR-047	US-016	UC-023	TBD
FR-048	US-016	UC-023	TBD

24. Requirements Validation

The SRS v1.0 was reviewed against the required quality criteria.

Validation Criterion	Review Result
Correctness	Each functional requirement is derived from the stakeholders, objectives or initial scope of Assignment 1. No initial-release requirement introduces ERP write-back, SSO, cross-university exchange or external regulatory approval.
Unambiguity	Functional requirements use a single, observable "The system shall..." behavior wherever practical. Roles, states and permitted actions are named explicitly.
Completeness	The requirements cover the full initial scope: access control, master data, proposal creation, documents, workflow, feedback, versioning, decisions, notifications, implementation tracking, search/dashboard/timeline and auditability.
Consistency	Role permissions, functional requirements and business rules were checked for contradictions. In particular, invited stakeholders can give feedback but cannot approve, and submitted versions are immutable while Draft/Returned proposals remain editable.
Verifiability	Functional requirements and selected user stories describe observable outcomes that can be checked through UI, API, database or audit-log tests. Quantified NFRs are included where practical.
Feasibility	The requirements are suitable for a student-developed web prototype using representative data and test adapters. They do not depend on unavailable production ERP, SSO or government systems.
Traceability	Every FR in this baseline is mapped to at least one user story and use case in the initial RTM. Test references are reserved as TBD for the testing phase.
Prioritization	Features are classified using MoSCoW and the Won't-Have list matches the agreed initial out-of-scope items.
Modifiability	Requirements are grouped by functional area and use stable IDs, allowing later changes to be made without renumbering unrelated requirements.

Validation Conclusion

The requirements are considered suitable as the **SRS v1.0 baseline** for subsequent UML analysis, system design, implementation and testing. Any later scope or requirement change should update the relevant FR/BR/NFR, affected user story and use case, MoSCoW priority and RTM entry.

25. Example of Requirement Quality

The quality of a requirement depends on whether a reviewer and tester can interpret it in only one reasonable way and verify its result.

Poor Requirement

The system should manage curriculum proposals quickly, securely and efficiently.

This statement is weak because "quickly", "securely" and "efficiently" are not independently testable, the actor and operation are unclear, and several concerns are mixed into one sentence.

Better Requirements

FR-015 - Validate Proposal Completeness

The system shall validate all mandatory fields and required supporting information before allowing submission.

NFR-001 - Performance

Under the expected prototype workload, at least 95% of normal authenticated page and CRUD requests should complete within 3 seconds, excluding file upload time.

NFR-005 - Security

Authorization shall be enforced on the server side for every protected operation, not only by hiding user-interface controls.

These statements are more useful because each specifies a concrete behavior or measurable quality that can be verified independently.

Additional Quality Examples

Poor: The system should notify everyone about important proposal changes.

This is ambiguous because "everyone" and "important" are undefined.

Better: FR-036 states that the system shall create an in-application notification for relevant stakeholders when a proposal is submitted, returned, revised, scheduled, approved, rejected or assigned for action. The triggering events and recipient category are therefore identifiable and testable.

Poor: Reviewers should not be able to do wrong actions.

This does not identify which actions are valid, at which stage they are valid, or how the system should respond.

Better: FR-024 states that the system shall prevent a proposal from bypassing a mandatory configured review stage unless the workflow configuration explicitly permits that transition. BR-010 further restricts formal review actions to users assigned to the responsible role or committee for the current stage.

Poor: Version history must be maintained properly.

The word "properly" is subjective.

Better: FR-025 and BR-005 require a new immutable version snapshot on submission or resubmission, while FR-026 and BR-006 require earlier submitted versions to remain viewable to authorized users. These requirements define the expected behavior precisely enough for test cases.

Requirement Review Checklist

Before accepting a new or modified requirement into the baseline, the team should ask:

- Is the actor or triggering condition clear?
- Is the system behavior stated as one observable outcome?
- Are important states, permissions and exceptions defined?
- Does the requirement duplicate or contradict another FR/BR/NFR?
- Can a test case determine pass or fail without subjective judgement?
- Is the requirement feasible within the agreed CS314 prototype scope?
- Is there a trace from the requirement to at least one stakeholder need, use case or user story?

26. Requirement Change Management

Because the project follows Agile - Iterative and Incremental Development, requirements may evolve after stakeholder review. Changes shall be controlled so that the SRS, backlog, design and tests remain consistent.

A typical change flow is:

Change Request -> Impact Analysis -> Requirement Update -> Backlog Update -> MoSCoW Re-prioritization -> Sprint Planning -> Design Update -> Implementation -> Testing -> Release -> Traceability Update

For every accepted change, the team should:

1. identify the stakeholder need and reason for change;
2. identify affected FRs, BRs, NFRs, data entities, use cases and user stories;
3. check whether the change expands the agreed initial scope;
4. estimate implementation and testing impact;
5. approve a new priority before development;
6. update requirement wording and preserve stable IDs where possible;
7. update UML/design artifacts and the product backlog;
8. add or update test cases;
9. update the Requirements Traceability Matrix;
10. record the SRS version in which the change became part of the baseline.

Example Change Request

CR-001 - Add email delivery for notifications.

The initial baseline requires in-application notifications. If access to an external email service becomes available, email delivery can be added as an additional channel. The change would primarily affect FR-036/FR-037, Notification data attributes, NFRs concerning reliability/security, relevant user stories, test cases and deployment configuration. It should not replace the in-application notification mechanism.

Initial Impact Assessment for CR-001

Area	Expected Impact
Functional Requirements	Extend notification delivery behavior while preserving in-application notifications.
Business Rules	Clarify which events and users are eligible for external notification.
Non-Functional Requirements	Add reliability, privacy and failure-handling expectations for email delivery.
Data Requirements	Add optional channel, delivery status, retry count and delivery timestamp to Notification data.
Use Cases / User Stories	Update notification and pending-action flows where email is a secondary channel.
Testing	Add tests for successful delivery, provider failure, duplicate prevention and unauthorized recipient prevention.
Deployment	Add a configured email provider or test adapter and protect provider credentials.
Traceability	Update the RTM and any affected acceptance criteria after the change is approved.

A change request shall not be treated as implemented merely because code exists. It becomes part of the controlled baseline only after the relevant requirement, design, test and traceability artifacts have been updated and the team has agreed on its priority.

27. Future Requirements

The following capabilities are potential future extensions and are not commitments for SRS v1.0 unless promoted through change management:

- institutional single sign-on when the institute provides authorized identity integration;
- direct integration with institute ERP, website or course-registration systems after formal API access is available;
- external email delivery and configurable reminder schedules in addition to in-application notifications;
- richer document comparison that highlights changes inside uploaded syllabus documents;
- calendar integration for committee meetings, proposal deadlines and implementation milestones;
- advanced turnaround-time and bottleneck analytics for departments and workflow stages;
- digital signing or stronger formal attestation of committee decisions if institutional policy requires it;
- automated checks for duplicate or closely related curriculum proposals;
- configurable archival and retention policies for old proposals and attachments;
- export of authorized proposal summaries, decision records and audit reports to PDF/CSV formats;
- optional rule-based impact checks for prerequisites, course dependencies and programme structures;
- cross-university proposal exchange only if a future inter-institutional data-sharing policy is established.

28. SRS Baseline and Versioning

- **Initial baseline:** SRS v1.0.
- **Baseline date:** 6 September 2026.
- **Baseline project:** Curriculum Revision Management System.
- **Baseline scope:** role-based access, master data, structured proposals, supporting documents, configured review workflow, stakeholder feedback, immutable version history, committee decisions, notifications, implementation tracking, search/filter/dashboard/timeline and auditability.
- Requirement IDs defined in this version should remain stable throughout the v1.x series. If a requirement is removed, its ID should be marked retired rather than silently reused for another behavior.
- Minor clarification that does not alter behavior may increment the document revision within v1.x.
- Addition, removal or behavioral change of a requirement shall be processed through Section 26 and reflected in the RTM.
- A future major release with materially different scope may establish SRS v2.0 after stakeholder approval.

Version Record

Version	Date	Status	Description
v1.0	6 September 2026	Initial Baseline	First complete requirements specification derived from Assignment 1 stakeholders, objectives and scope.
v1.x	TBD	Controlled Revision	Reserved for approved clarifications or requirement changes that remain within the same project baseline.
v2.0	TBD	Future Major Baseline	Reserved for a materially expanded or re-baselined project scope after stakeholder approval.

Any future revision should include a short change summary and identify the affected requirement IDs so that reviewers can understand what changed without comparing the complete document manually.

29. Individual Contribution

Team Member	Roll Number	Assigned Contribution / Responsibility
Utkarsh	24UCC096	Requirements & Analysis Lead - organized the user classes; drafted core proposal, review and access-control functional requirements; contributed business rules and performed requirements-validation review.
Piyush	24UCC097	Workflow & Architecture Lead - prepared the system-context and proposal-lifecycle modelling; contributed workflow, versioning and data requirements; documented workflow-oriented detailed use-case behavior.
Yash	24UCC099	Documentation & Traceability Lead - organized the major use-case catalogue and user stories; prepared acceptance criteria, MoSCoW prioritization, traceability mapping, document consistency and SRS baseline/versioning.
Mehul	24UCC100	Quality & Non-Functional Requirements Lead - drafted performance, security, usability, reliability and auditability requirements; contributed requirement-quality examples, change-management rules, future-requirement review and final quality checks.

The contribution division above represents the team's planned/documented ownership for this requirements baseline. All members are expected to review the complete SRS and be able to explain every submitted section during evaluation or viva.

AI USAGE POLICY - STUDENT DECLARATION

AI tools may be used as supporting tools for this assignment. The team is responsible for the correctness, consistency and understanding of all submitted requirements.

- We have disclosed the use of AI tools, if any.
- AI-generated suggestions have been reviewed, verified and adapted by the team.
- We understand and can explain the submitted requirements, use cases and user stories.
- We have verified that the requirements are consistent with our project's scope.
- The final SRS represents our team's understanding and decisions.

Item	Details
AI Tool(s) Used	ChatGPT
Purpose of AI Use	Assistance in structuring the requirements specification, drafting functional/non-functional requirements, use cases, user stories, acceptance criteria and traceability, and improving clarity of language. All project decisions and final content were reviewed and verified by the team.
Team ID	T017
Team Representative	YASH GUPTA
Date	6 September 2026

The LNM Institute of Information Technology, Jaipur
Software Development Laboratory
Course Code: CS314

Lab Assignment 3: Scenario, Use Case and Activity Modelling

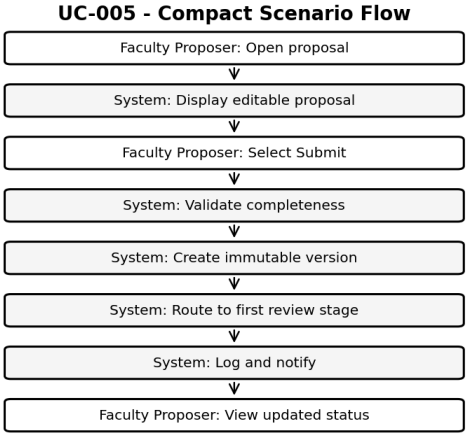
Project: Curriculum Revision Management System

Team ID	T017
Team Members	Utkarsh, Piyush, Yash, Mehul
Roll Numbers	24UCC096, 24UCC097, 24UCC099, 24UCC100
Development Model	Agile - Iterative and Incremental Development
Date	12 September 2026

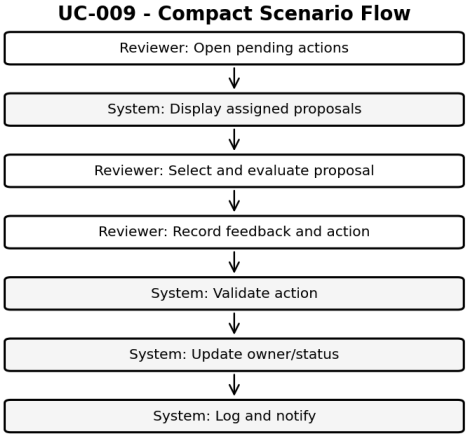
30. Scenarios

The same three detailed use cases defined in Assignment 2 are carried forward without introducing new functionality: UC-005 Submit Proposal, UC-009 Review Proposal and Record Feedback, and UC-015 Record Committee Decision.

Scenario: UC-005 - Submit Proposal

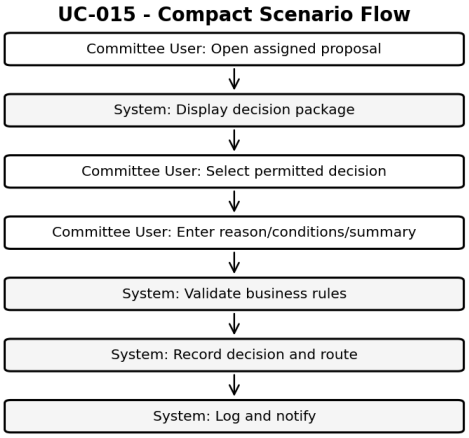
Actor(s) involved	Primary: Faculty Proposer. Supporting: Curriculum Revision Management System.
Starting condition	The Faculty Proposer is authenticated, owns the proposal, the proposal is in Draft or Returned for Revision status, and the applicable review workflow is configured and active.
Main sequence of activities	1. The proposer opens the proposal. 2. The system displays the current editable proposal data and supporting documents. 3. The proposer selects Submit. 4. The system validates all mandatory fields and required information for the selected proposal type. 5. The system creates an immutable proposal-version snapshot. 6. The system changes the proposal to Submitted / Under Review. 7. The system determines the first configured review stage and responsible role. 8. The system assigns the pending action to that stage. 9. The system records submission and version events in the timeline and audit trail. 10. The system generates relevant in-application notifications. 11. The system displays confirmation and the updated current status.
Important alternative / exception flow	• Mandatory data missing: submission is blocked and the missing or invalid fields are identified. • Invalid supporting document: the file is rejected and the proposal remains editable. • Workflow unavailable: submission is blocked and the system reports that workflow configuration is required. • Unauthorized user: a user who does not own the editable proposal is denied the submission action.
Expected outcome	A new immutable submitted version exists; the proposal is assigned to the first configured review stage; the submitted version is no longer editable; and the new status, pending action, timeline/audit entries and notification are available to authorized users.
Compact representation	

Scenario: UC-009 - Review Proposal and Record Feedback

Actor(s) involved	Primary: Authorized Reviewer / Committee Member. Supporting: Curriculum Revision Management System; Faculty Proposer or next responsible role receives the resulting notification.
Starting condition	The reviewer is authenticated, is permitted to act at the proposal's current workflow stage, the proposal has at least one submitted version, and the proposal is assigned to the reviewer's role or committee.
Main sequence of activities	1. The reviewer opens the pending-action list. 2. The system displays proposals assigned to the reviewer's role. 3. The reviewer selects a proposal. 4. The system displays the current proposal version, supporting documents, previous review context and relevant version history. 5. The reviewer evaluates the proposal. 6. The reviewer records a comment, recommendation or requested change. 7. The reviewer selects a permitted workflow action such as Recommend/Forward or Return for Revision. 8. The system validates that the selected action is permitted at the current stage. 9. If the action is Return for Revision, the system requires a reason. 10. The system records the review action with reviewer identity and timestamp. 11. The system updates proposal status/current owner according to the workflow. 12. The system creates timeline, audit and notification events.
Important alternative / exception flow	• Reviewer no longer authorized: the action is denied. • Stale proposal stage: the action is blocked and the current status is refreshed. • Missing return reason: Return for Revision cannot be completed until a reason is entered. • Comment-only review: when permitted, a contextual comment may be saved without advancing the workflow.
Expected outcome	The review action is stored with actor and timestamp; the proposal remains at the current stage or moves to the configured next owner; the proposer or next responsible role is notified; and the action appears in the proposal timeline and audit trail.
Compact representation	

Scenario: UC-015 - Record Committee Decision

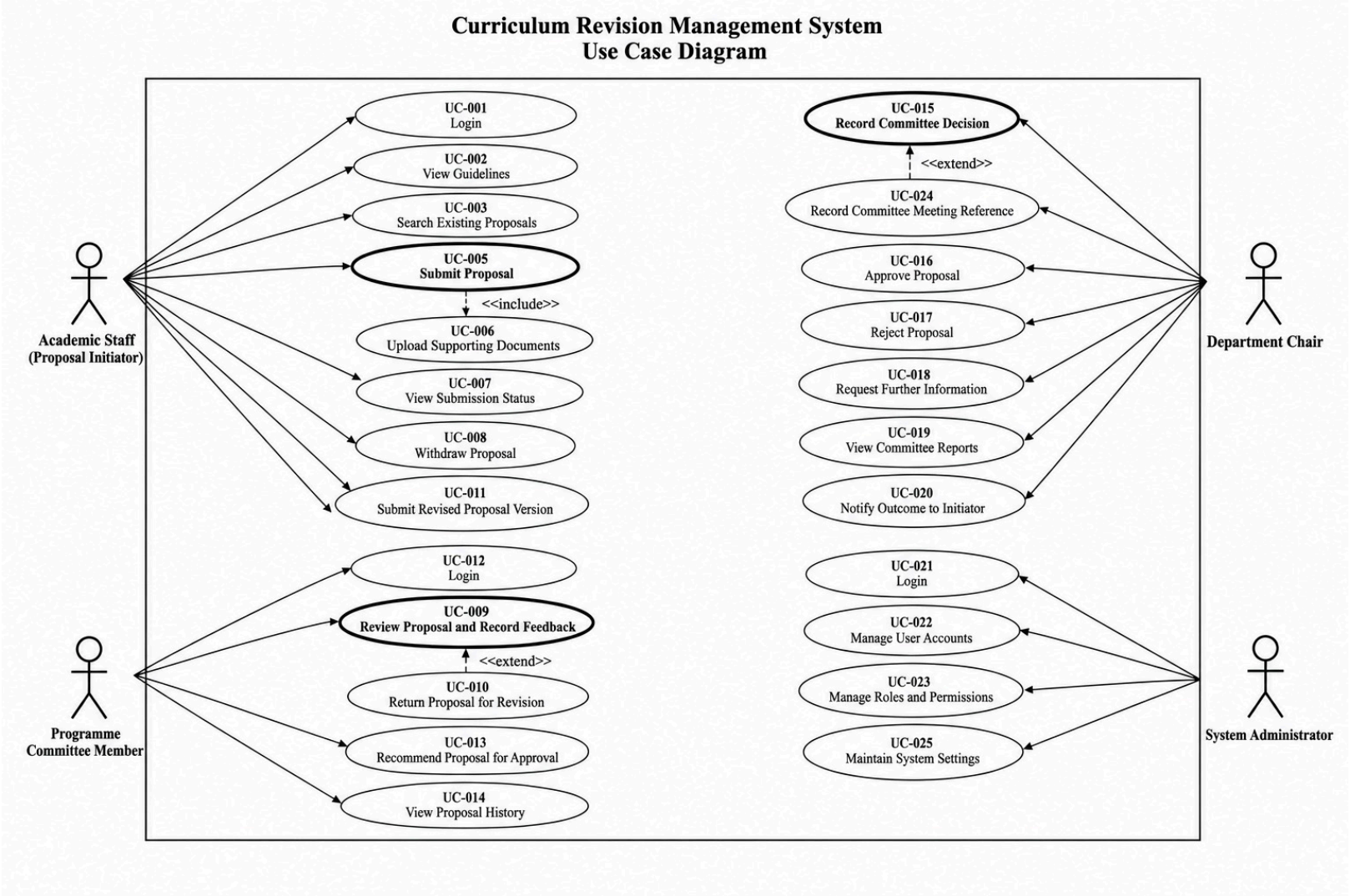
Actor(s) involved	Primary: Authorized Committee User. Supporting: Curriculum Revision Management System; Academic Office and next responsible authority are downstream recipients when applicable.
Starting condition	The committee user is authenticated; the proposal has reached a workflow stage at which the committee is authorized to decide; the current submitted version is available; and meeting context is available when required by the configured process.
Main sequence of activities	1. The committee user opens the proposal assigned for decision. 2. The system displays the current version, supporting documents, prior recommendations and relevant feedback. 3. The committee user associates the proposal with a meeting reference if required. 4. The committee user selects a permitted decision: Recommendation, Conditional Approval, Return or Rejection. 5. The committee user enters the required reason, conditions or recommendation/voting summary. 6. The system validates that the decision and supporting fields satisfy the business rules. 7. The system records the decision with the proposal version, actor and timestamp. 8. The system updates the workflow to the configured next stage or final state.

	9. The system creates timeline and audit events. 10. The system notifies the proposer, Academic Office and next responsible role as applicable.
Important alternative / exception flow	• Missing reason: Return or Rejection cannot be saved until a reason is recorded. • Missing conditions: Conditional Approval cannot be saved until at least one condition is recorded. • Unauthorized decision type: a decision not permitted for the current committee/stage is blocked. • Higher authority required: the committee outcome is recorded as a recommendation and routed onward instead of being marked finally approved. • Proposal already moved: duplicate decision recording is prevented.
Expected outcome	The formal committee decision is stored against the relevant proposal version, is visible in authorized proposal history and audit records, and the proposal is routed to the next configured stage or reaches the appropriate final state.
Compact representation	

31. UML Use Case Diagram - Complete System

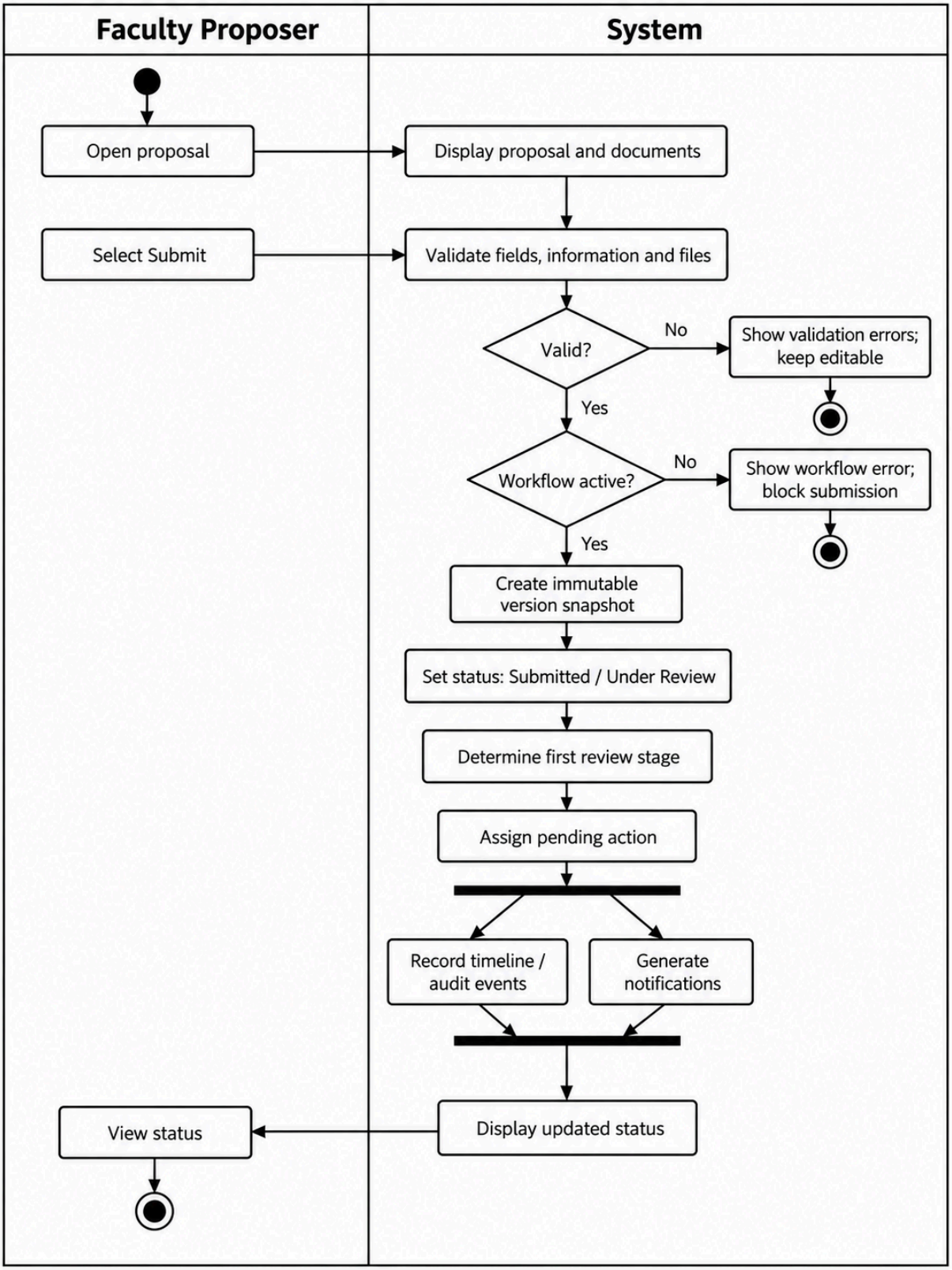
The complete use-case model below contains the 25 use cases already defined in Assignment 2. The three carried-forward detailed use cases are highlighted. Optional review/meeting behavior is shown with <<extend>>, and UC-011 is modeled as a specialized resubmission form of UC-005.

Relationship notes: UC-010 Return Proposal for Revision extends UC-009 Review Proposal and Record Feedback; UC-024 Record Committee Meeting Reference extends UC-015 when meeting context is required; UC-011 Submit Revised Proposal Version specializes the core submission behavior represented by UC-005.

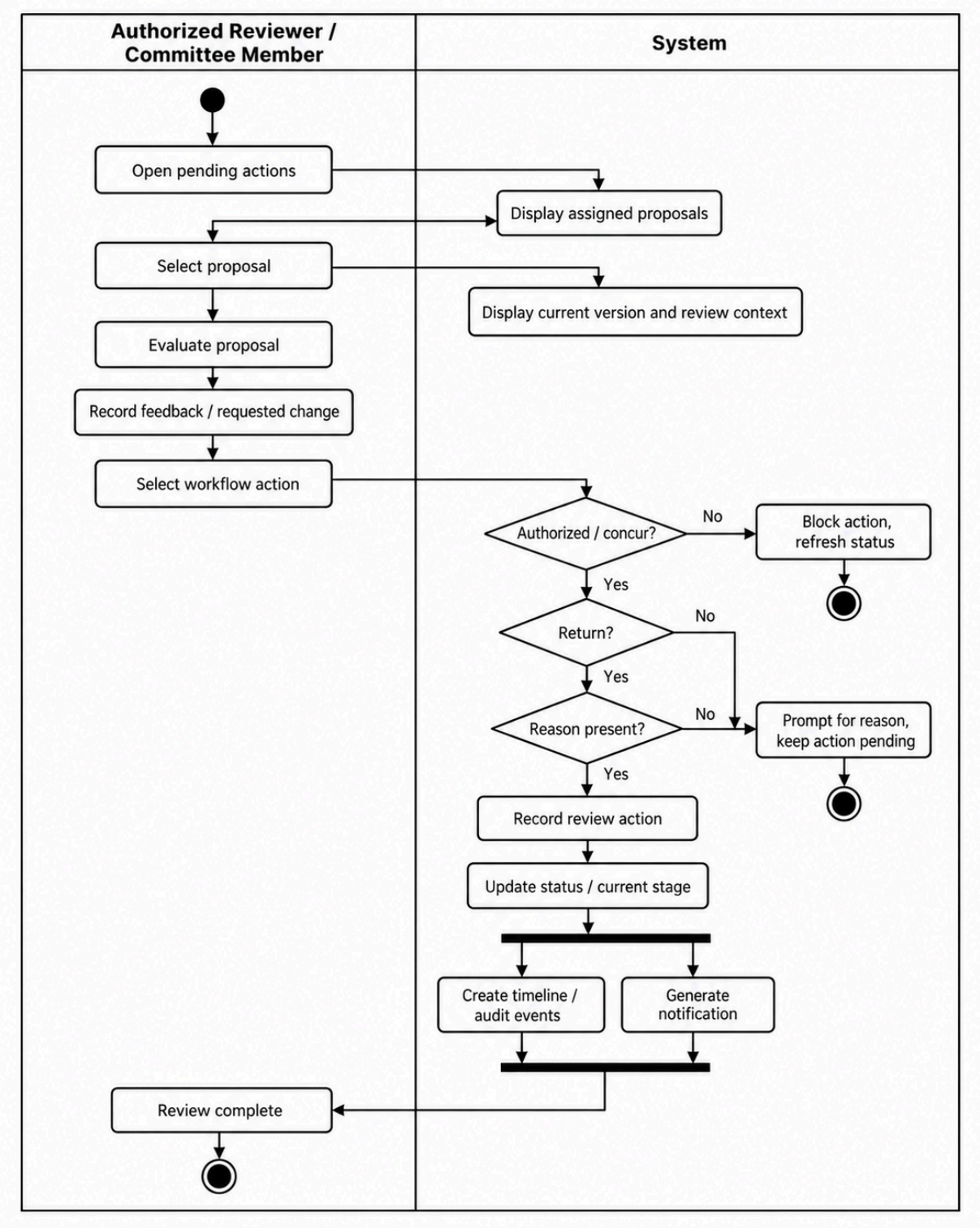


32. Activity Diagrams

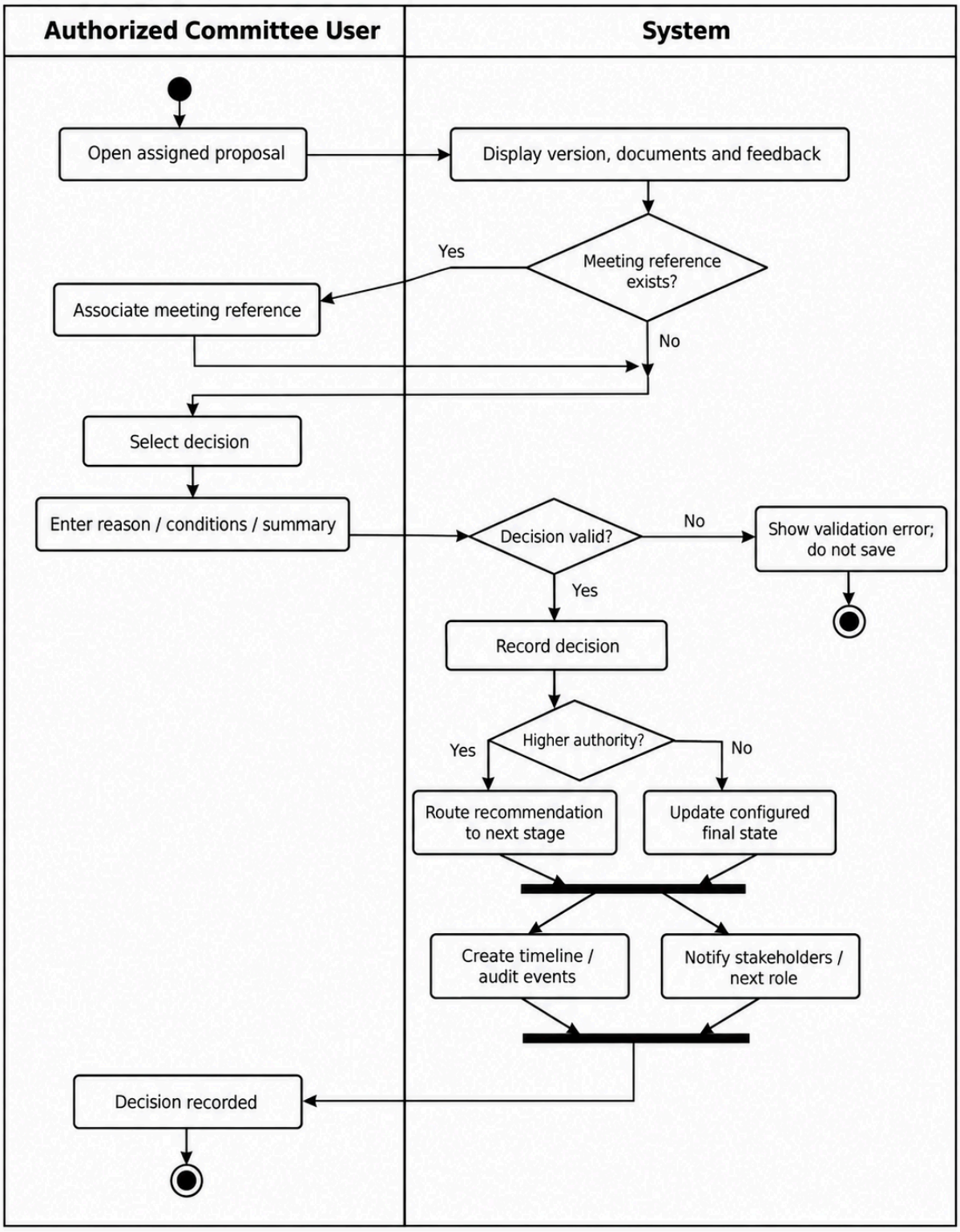
32.1 UC-005 - Submit Proposal



32.2 UC-009 - Review Proposal and Record Feedback



32.3 UC-015 - Record Committee Decision



33. Consistency and Traceability Check

The models preserve the required chain: Assignment 2 requirements -> the same 3 detailed use cases -> scenarios -> complete-system use case diagram -> activity diagrams. No new system functionality has been introduced.

Detailed Use Case	Assignment 2 Source	Requirement Basis	Scenario	Activity Diagram
UC-005 Submit Proposal	Section 19.1	FR-015, FR-016, FR-018; versioning, notification and audit behavior already defined in the SRS	Scenario 1	32.1
UC-009 Review Proposal and Record Feedback	Section 19.2	FR-019 to FR-024; notification and audit behavior already defined in the SRS	Scenario 2	32.2
UC-015 Record Committee Decision	Section 19.3	FR-031 to FR-034; notification and audit behavior already defined in the SRS	Scenario 3	32.3

35. Individual Contribution

• **Utkarsh (24UCC096) -** Prepared and verified the three scenarios against the detailed use cases and checked requirements-to-model consistency.

• **Piyush (24UCC097) -** Designed the complete-system UML use case model and reviewed actor/use-case associations and UML relationships.

• **Yash (24UCC099) -** Prepared the UC-005 and UC-009 activity models and organized the assignment documentation and traceability presentation.

• **Mehul (24UCC100) -** Prepared the UC-015 activity model, reviewed alternative/exception paths, and performed final UML and document quality checks.

AI USAGE POLICY - STUDENT DECLARATION

AI tools may be used as supporting tools for this assignment. The team is responsible for the correctness, consistency and understanding of all submitted scenarios and UML models.

- We have disclosed the use of AI tools, if any.
- AI-generated suggestions have been reviewed, verified and adapted by the team.
- We understand and can explain the submitted scenarios, use case diagram and activity diagrams.
- We have verified that the UML models are consistent with our project's requirements, use cases and scope.
- The final submission represents our team's understanding and modeling decisions.

AI Tool(s) Used	ChatGPT
Purpose of AI Use	Assistance in structuring scenarios and UML models, checking consistency with Assignment 2, and improving clarity. All final modeling decisions and content were reviewed by the team.
Team ID	T017
Team Representative	YASH GUPTA
Date	12 September 2026

The LNM Institute of Information Technology, Jaipur
Software Development Laboratory
Course Code: CS314

Lab Assignment 4: Classes, Object Diagram and Relationships

36. Classes

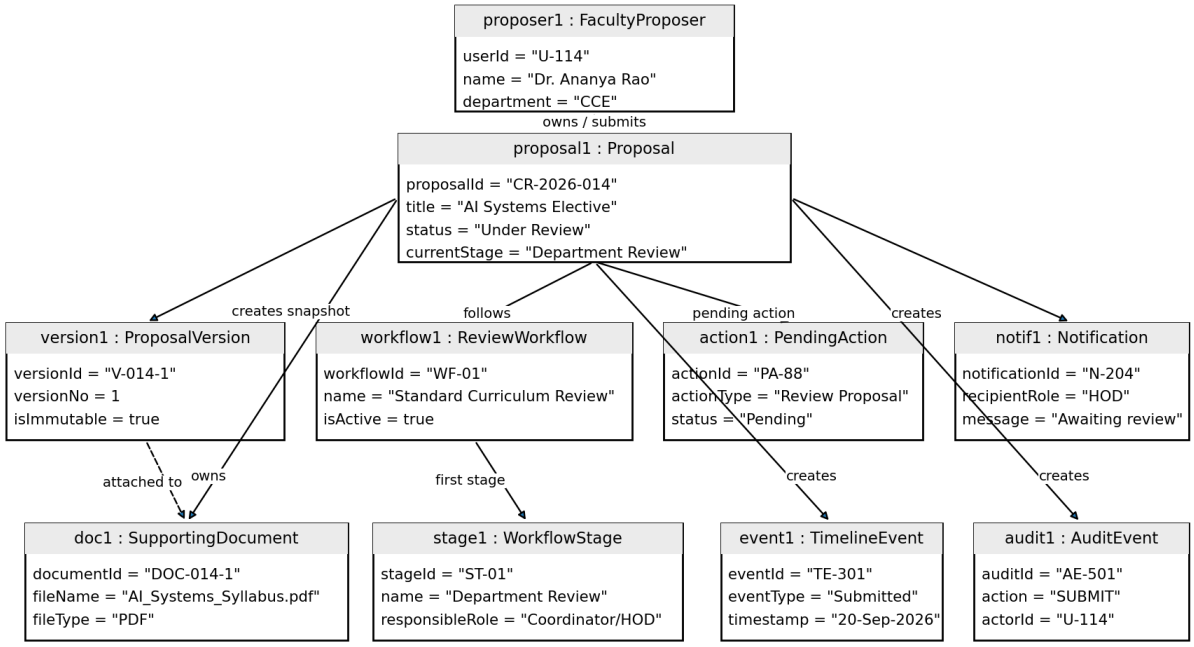
Class	Key Attributes	Key Operations	Traces to
User (abstract)	userId, name, email, passwordHash, roles	login(), logout(), hasRole()	FR-001-006
FacultyProposer	employeeId, department	submitProposal(), reviseProposal()	FR-015-016, UC-005
Reviewer	reviewerId, reviewRole	reviewProposal(), recordFeedback(), forwardProposal()	FR-019-024, UC-009
CommitteeMember	committeeRole	recordDecision(), associateMeeting()	FR-031-034, UC-015
Proposal	proposalId, title, proposalType, status, currentStage, ownerId	validateCompleteness(), submit(), changeStatus()	FR-015-024, BR-003-004
ProposalVersion	versionId, versionNo, createdAt, createdBy, isImmutable	createSnapshot(), compareTo()	FR-025-027, BR-005-006
SupportingDocument	documentId, fileName, fileType, fileSize	validateFile(), attach()	FR-013, UC-006, SRS §16.10
ReviewWorkflow	workflowId, name, isActive	getFirstStage(), getNextStage(), isActionAllowed()	FR-008, FR-018, FR-024
WorkflowStage	stageId, name, sequenceNo, responsibleRole	validateAction(), nextStage()	FR-018, FR-024, BR-007/010
PendingAction	actionId, actionType, status, assignedRole, createdAt	assign(), complete()	FR-017-018, FR-037

Class	Key Attributes	Key Operations	Traces to
ReviewFeedback	feedbackId, type, comment, reason, timestamp	record(), validateReturnReason()	FR-020-022, BR-008/010
Committee	committeeId, name, committeeType	addMember(), scheduleMeeting()	FR-031-034
CommitteeMeeting	meetingId, meetingRef, meetingDate	associateProposal()	FR-031, UC-015
CommitteeDecision	decisionId, decisionType, reason, conditions, votingSummary, decisionDate	validateDecision(), record()	FR-032-034, BR-008/009/014
Notification	notificationId, message, recipientId, createdAt	send()	FR-036, UC-005/009/015
TimelineEvent	eventId, eventType, description, timestamp	record()	FR-045, UC-005/009/015
AuditEvent	auditId, action, actorId, objectId, timestamp	record()	FR-047, UC-005/009/015
ImplementationMilestone	milestoneId, title, targetDate, status	update(), attachEvidence()	FR-038-041, US-013, UC-019, SRS §16.19

SRS data-model naming note: Proposal corresponds to CurriculumProposal (SRS §16.8), ReviewWorkflow to WorkflowDefinition (§16.11), ReviewFeedback to Feedback (§16.15), and CommitteeDecision to Decision (§16.17).

37. Object Diagram

UC-005 (Submit Proposal).



Description: Dr. Ananya Rao (proposer1) submits the curriculum revision proposal CR-2026-014 for a new AI Systems elective. The system validates proposal1, creates immutable version1, and moves the proposal to the first active workflow stage, Department Review. The supporting syllabus document doc1 remains owned by proposal1 and is linked to version1 as the submitted version; action1 is assigned to the Programme Coordinator/HOD. notif1, event1 and audit1 record the notification, timeline and audit consequences of the submission. This is the snapshot immediately after the main flow of UC-005; every object shown is an instance of a class identified in Section 36.

38. Relationships

Generalization / Inheritance

- FacultyProposer and Reviewer specialize User (inherit from the general parent class User) because both use the same authenticated user identity and role-based access behavior (FR-001-006).
- CommitteeMember specializes Reviewer because a committee member can review a proposal and additionally record formal committee decisions in UC-015.

Composition

- Proposal "1" *-- "0..*" ProposalVersion - each submitted/resubmitted snapshot belongs to one proposal and remains part of its permanent version history (FR-025, BR-005-006).
- Proposal "1" *-- "0..*" SupportingDocument - each uploaded document is owned by its proposal, including documents attached while the proposal is still a draft (FR-013, UC-006, SRS §16.10).
- ReviewWorkflow "1" *-- "1..*" WorkflowStage - configured stages are parts of a particular workflow and define its ordered review sequence (FR-008, FR-024).
- ProposalVersion "1" *-- "0..*" ReviewFeedback - review comments and requested changes are stored in the context of the proposal version being reviewed (FR-020, FR-030).
- Proposal "1" *-- "0..*" ImplementationMilestone - implementation milestones are lifecycle records of an approved proposal and support status/evidence tracking (FR-038-041, US-013, UC-019).

Aggregation

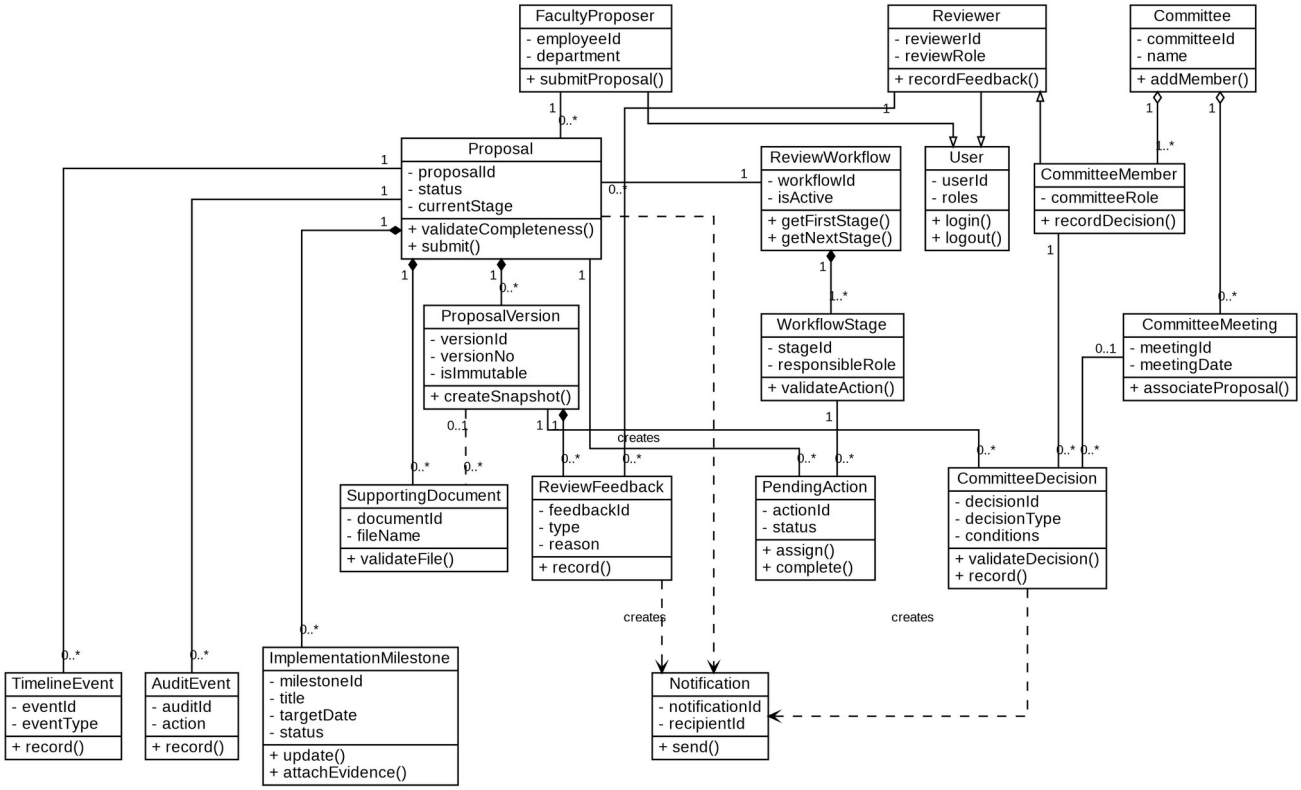
- Committee "1" o-- "1..*" CommitteeMember - members are grouped in a committee but continue to exist as users/reviewers independently of that committee.
- Committee "1" o-- "0..*" CommitteeMeeting - a committee maintains meeting records, while each meeting remains a separately identifiable record.

Association

- FacultyProposer "1" -- "0..*" Proposal : owns/submits (UC-005).
- Proposal "0..*" -- "1" ReviewWorkflow : follows the active configured workflow (FR-018, FR-024).
- WorkflowStage "1" -- "0..*" PendingAction : assigns work to the responsible role.
- Proposal "1" -- "0..*" PendingAction : has review actions over its lifecycle.
- Reviewer "1" -- "0..*" ReviewFeedback : writes comments, recommendations or requested changes (FR-020).
- CommitteeMember "1" -- "0..*" CommitteeDecision : records formal decisions (FR-032-034).
- CommitteeMeeting "0..1" -- "0..*" CommitteeDecision : provides meeting context when required (FR-031).
- ProposalVersion "1" -- "0..*" CommitteeDecision : each formal decision is stored against the relevant version.
- ProposalVersion "0..1" -- "0..*" SupportingDocument : optionally records the submitted version associated with a document; a draft document may have no version yet (FR-013, UC-006, SRS §16.10).
- Proposal "1" -- "0..*" TimelineEvent : keeps the chronological proposal history (FR-045).
- Proposal "1" -- "0..*" AuditEvent : records significant proposal actions with actor and timestamp (FR-047).

Dependency

- Proposal ..> Notification <<creates>> - successful submission triggers notifications to the responsible review role (UC-005, FR-036).
- ReviewFeedback ..> Notification <<creates>> - a completed review action notifies the proposer or next responsible role (UC-009, FR-036).
- CommitteeDecision ..> Notification <<creates>> - a recorded decision notifies the proposer, Academic Office and next responsible role as applicable (UC-015, FR-036).



39. Individual Contribution

- Utkarsh (24UCC096) - Identified the classes and traced them to the three detailed use cases and corresponding requirements.
- Piyush (24UCC097) - Prepared the UML Class Diagram and verified multiplicities and inheritance relationships.
- Yash (24UCC099) - Prepared the Object Diagram for UC-005 and wrote the snapshot description.
- Mehul (24UCC100) - Documented and justified the class relationships and performed the final consistency check.

AI USAGE POLICY - STUDENT DECLARATION

AI tools may be used as supporting tools for this assignment. The team is responsible for the correctness, consistency and understanding of all submitted classes, objects and relationships.

- We have disclosed the use of AI tools, if any.
- AI-generated suggestions have been reviewed, verified and adapted by the team.
- We understand and can explain the submitted classes, object diagram and relationships.
- We have verified that the UML models are consistent with our project's requirements, use cases and scope.
- The final submission represents our team's understanding and modeling decisions.

AI Tool(s) Used: ChatGPT

Purpose of AI Use: Used for structuring the assignment, checking consistency with Assignments 2 and 3, and drafting the UML diagrams. All final modeling decisions were reviewed by the team.

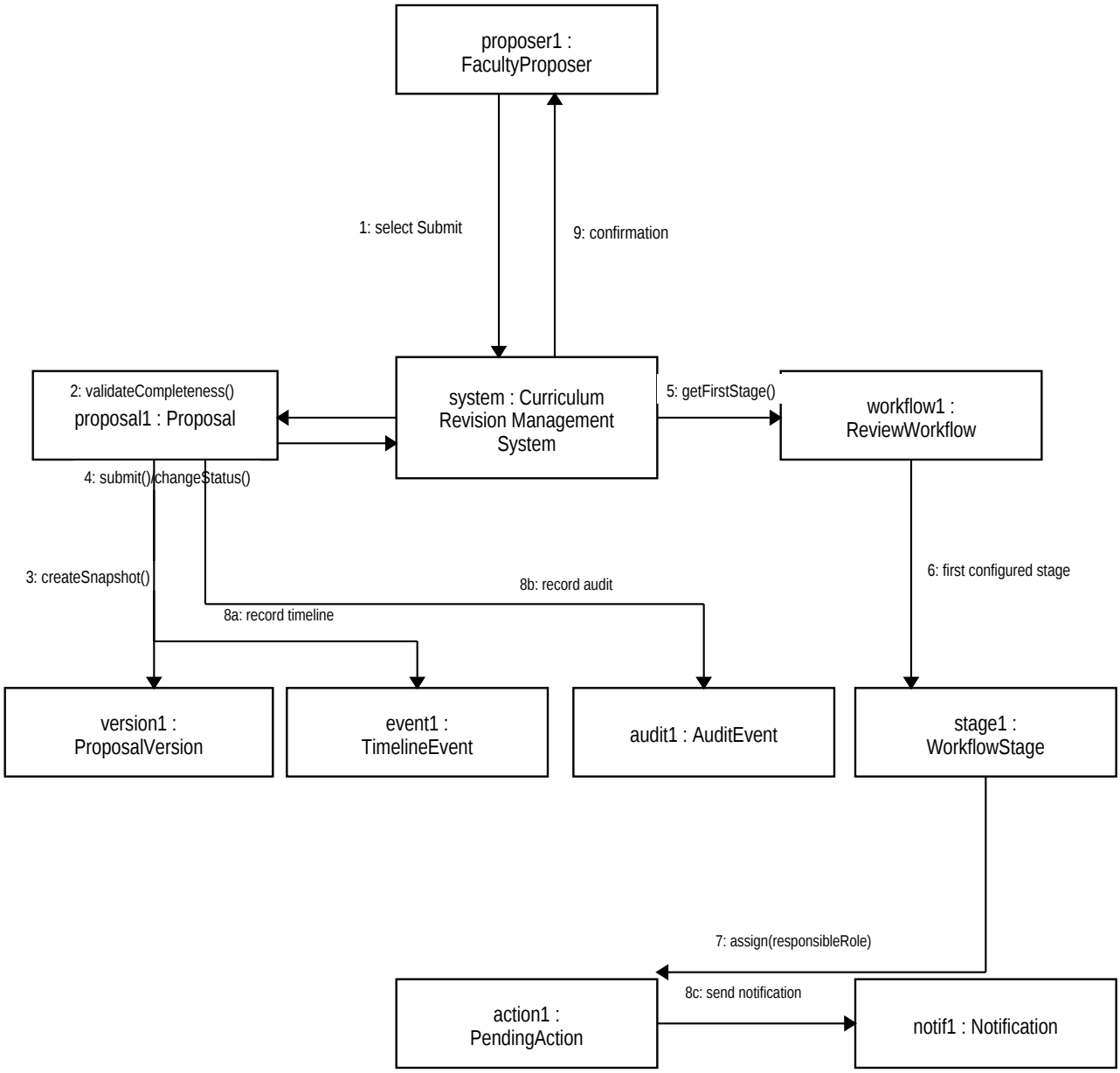
Team ID: T017

Team Representative: YASH GUPTA

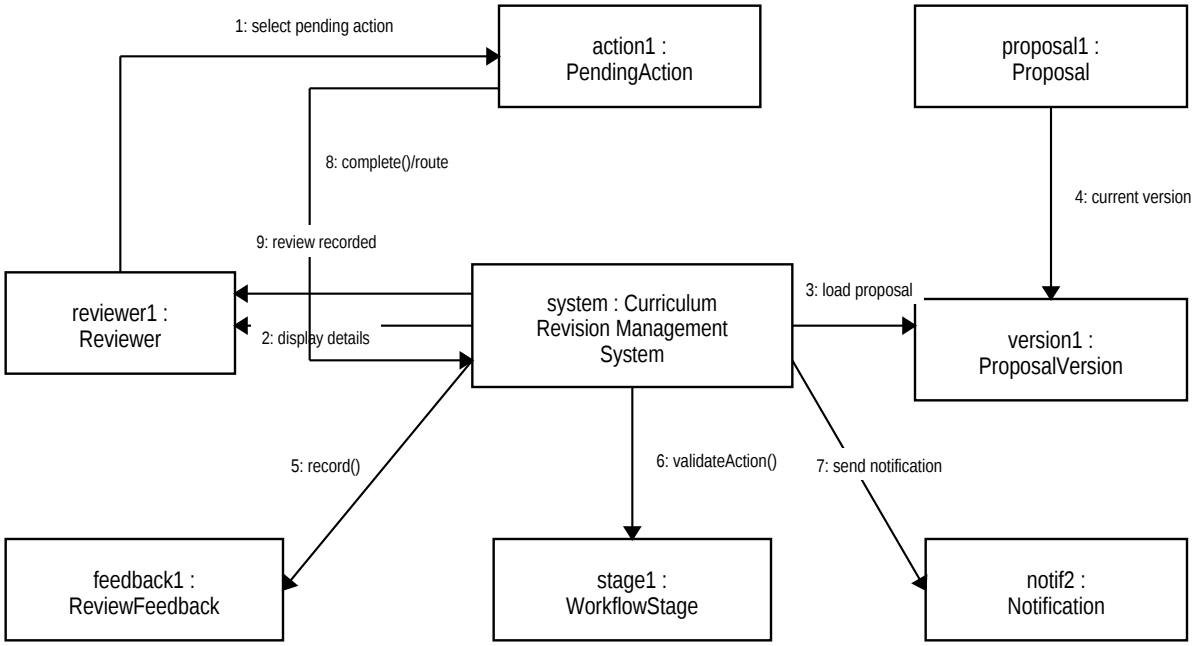
Date: 20 September 2026

40. Collaboration Diagrams

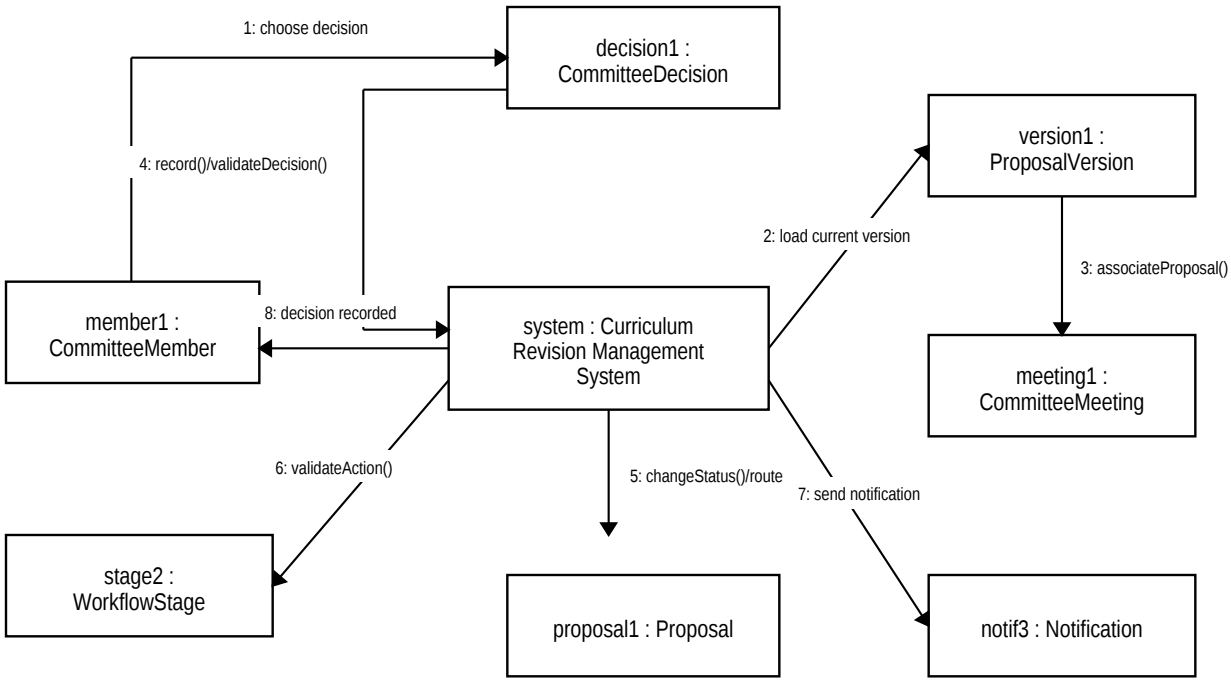
40.1 UC-005 - Submit Proposal



40.2 UC-009 - Review Proposal and Record Feedback

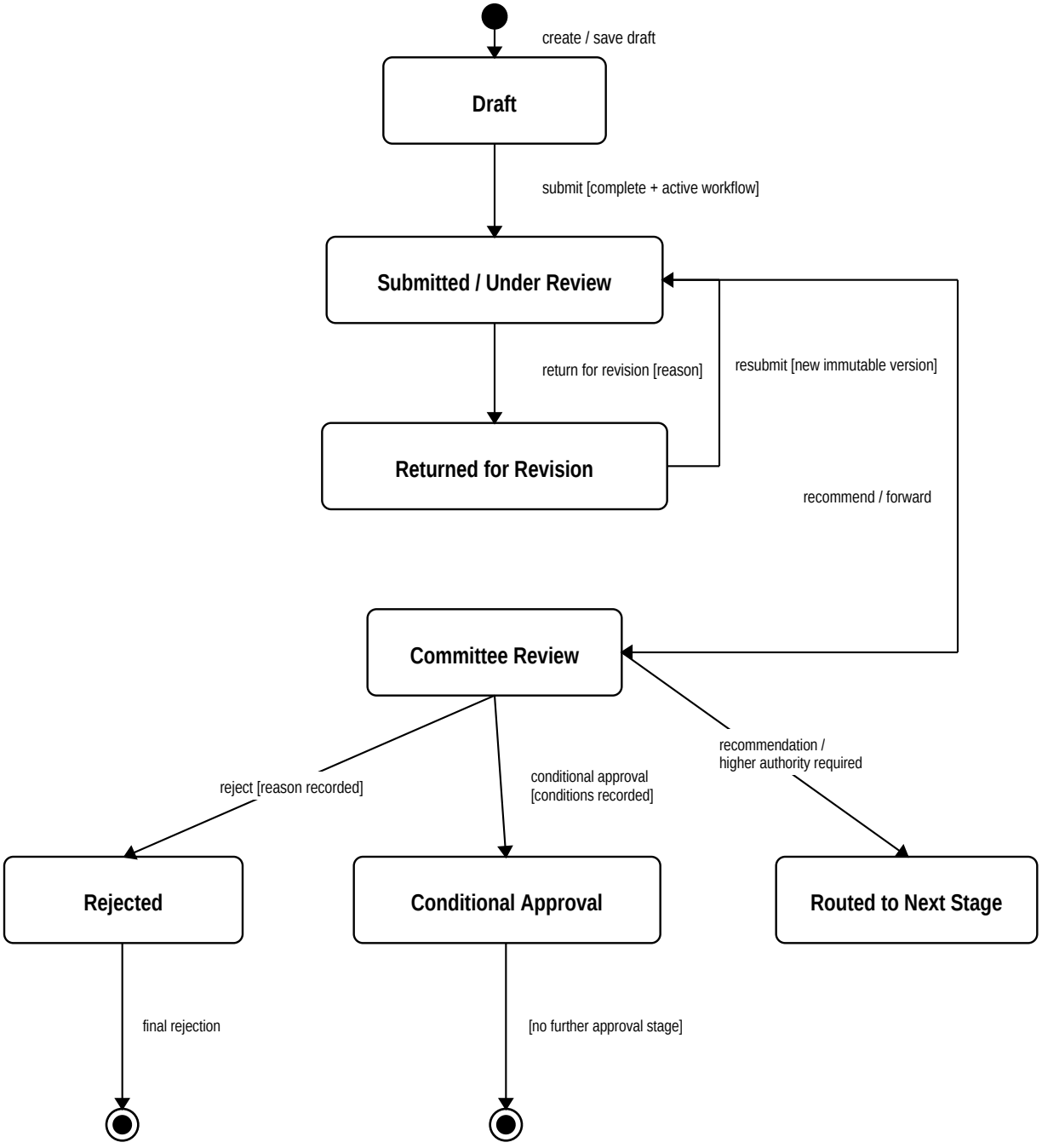


40.3 UC-015 - Record Committee Decision

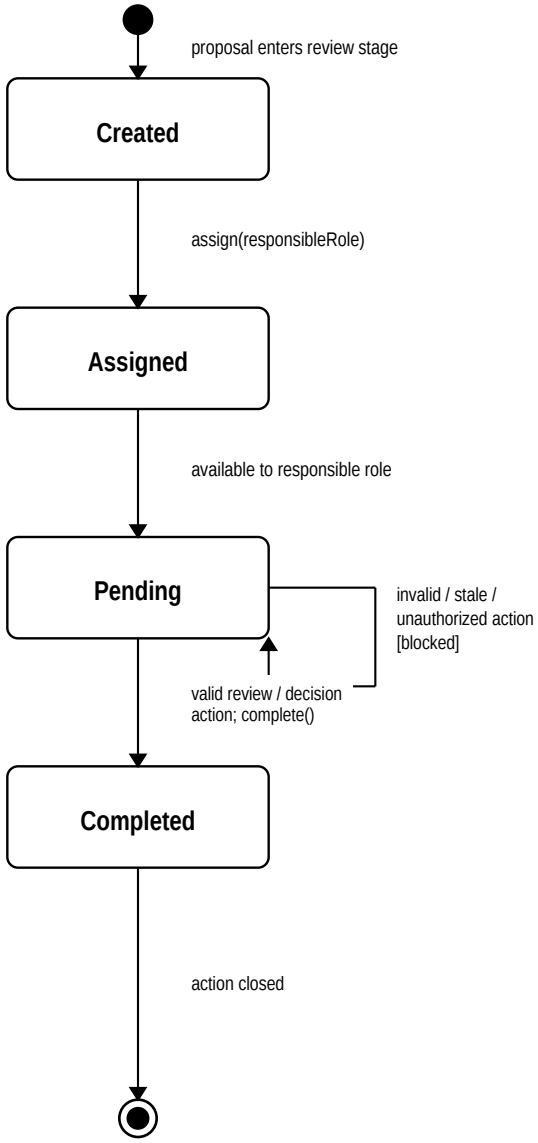


41. State Diagrams

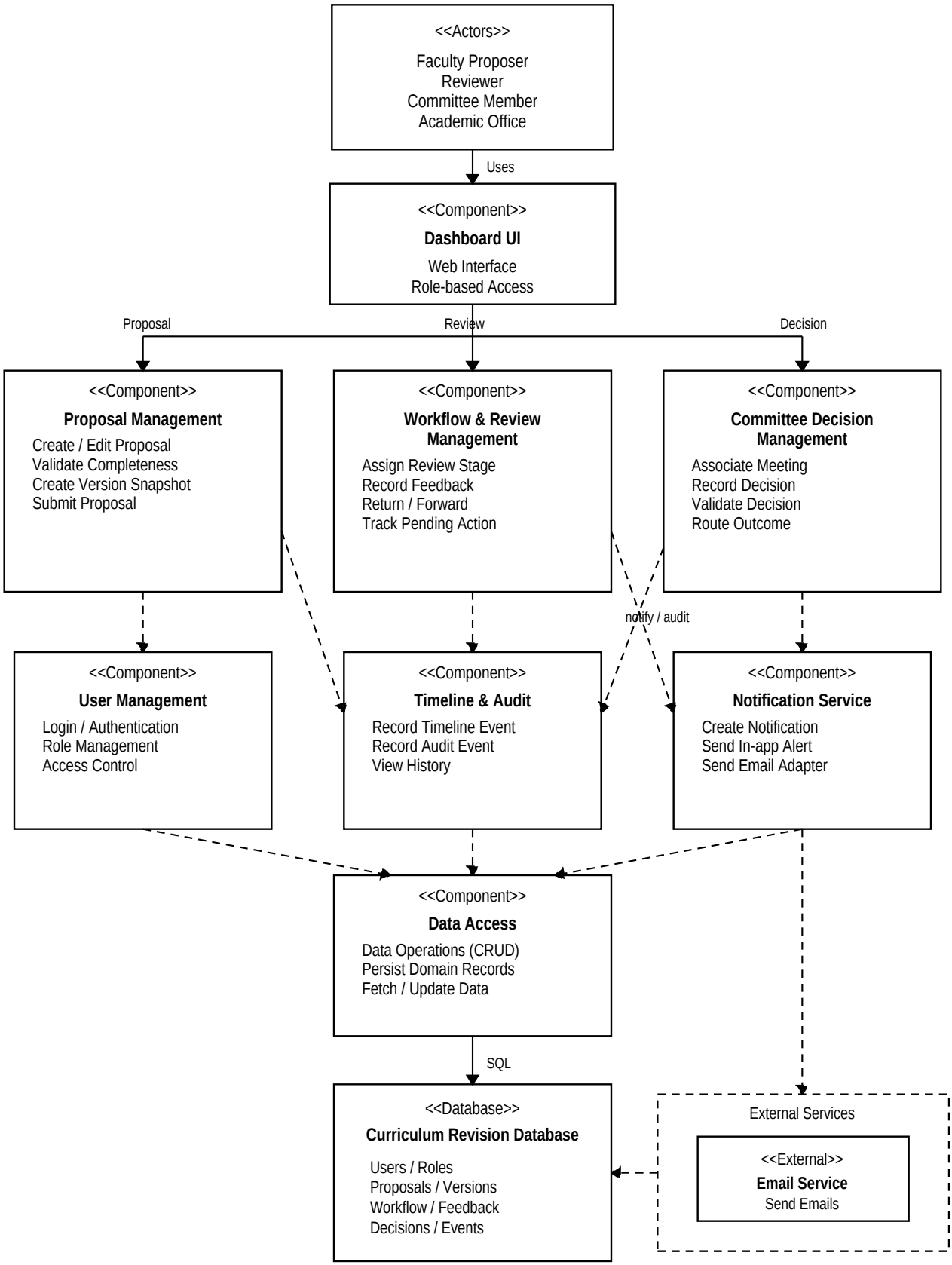
41.1 Proposal Object - UC-005 / UC-009 / UC-015



41.2 PendingAction Object - UC-005 / UC-009 / UC-015



42. Component Diagram



43. Individual Contribution

- Utkarsh (24UCC096) - Prepared and reviewed the UC-005 collaboration diagram and checked consistency with Assignment 4 classes.
- Piyush (24UCC097) - Prepared the UC-009 collaboration diagram and the Proposal state diagram.
- Yash (24UCC099) - Prepared the UC-015 collaboration diagram and the component diagram.
- Mehul (24UCC100) - Prepared the PendingAction state diagram and performed final UML consistency and presentation checks.

AI USAGE POLICY - STUDENT DECLARATION

AI tools may be used as supporting tools for this assignment. The team is responsible for the correctness, consistency and understanding of all submitted collaboration, state and component models.

- We have disclosed the use of AI tools, if any.
- AI-generated suggestions have been reviewed, verified and adapted by the team.
- We understand and can explain the submitted collaboration diagrams, state diagrams and component diagram.
- We have verified that the UML models are consistent with our project requirements, use cases, classes, objects and scope.
- The final submission represents our team understanding and modeling decisions.

AI Tool(s) Used	ChatGPT
Purpose of AI Use	Used for structuring the assignment, drafting UML diagrams, and checking consistency with Assignments 2 to 4. Final diagrams were reviewed by the team.
Team ID	T017
Team Representative	YASH GUPTA
Date	29 September 2026